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QUANTUM THREAT TIMELINE REPORT 2025

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Contents

Summary	4
1 Introduction	9
1.1 Quantum computing	9
1.2 Quantum threat to cybersecurity	10
1.3 Quantum computing towards fault tolerance	13
1.3.1 From NISQ to FASQ	14
2 Scope of this report.....	17
2.1 Related literature	17
3 Participants	19
4 Survey results.....	21
4.1 Physical realizations	22
4.2 Quantum factoring.....	24
4.2.1 Coarse-grained likelihood estimates.....	30
4.2.2 Comparison with previous years.....	32
4.3 Most promising scheme for fault-tolerance	37
4.4 Useful applications of early quantum computers.....	39
4.5 `Under the radar` research	41
4.6 Societal and funding factors	43
4.6.1 Level of funding of quantum computing research.....	43
4.6.2 Global race to build a fault-tolerant quantum computer	45
4.6.3 Ethical concerns	47
4.7 Current progress	49
4.7.1 Recent developments	49
4.7.2 Next near-term step.....	49
4.8 Sources of unexpected speed-up.....	51
Summary and outlook.....	53
References	56
A. Appendix.....	59
A.1 List of respondents	59
A.2 Realizations of quantum computers.....	63

Physical realizations	63
Models of computation	64
Error correction, fault tolerance, and logical qubits.....	65
Examples of error correcting codes	66
A.3 Survey Questions	66
Questions about “Implementations of quantum computing”	68
Questions about “Timeframe estimates”	68
Questions on “Other factors that may impact the quantum threat timeline”	68
Questions on “Current progress in the development of a cryptographically relevant quantum computer”	69
A.4 Responses and analysis.....	70
Quantum factoring responses and analysis	70
A.5 Glossary	73

Declaration on Potential Conflict of Interest

evolutionQ Inc. offers services and quantum-safe cybersecurity products to help clients deploy and manage quantum-safe technologies across their networks.

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Summary

Several widely used cybersecurity protocols rely on computational challenges believed to be practically unsolvable by classical computers. The advent of quantum computers that exploit quantum phenomena at the microscopic level in a highly controlled manner would allow some of those challenges to be overcome, posing severe cybersecurity risks.

To address this threat, new classical and quantum cryptographic techniques that are proven or considered to be resistant to quantum attacks can be employed. Unfortunately, upgrading cryptosystems to *quantum-safe cryptography* is a complex process. It involves developing and implementing new hardware and software, establishing standards, and updating older systems, all while maintaining a functioning cryptographic infrastructure and managing both past and current sensitive data.

Because of the complexity and time required by such a process, a successful transition cannot take the form of reactive, rushed crisis management; it should be made an integral part of a proactive technology lifecycle

management. The urgency of moving to quantum-resistant cryptography varies for each organization, based on its security needs and risk tolerance. This urgency can be gauged using three primary factors:

- the *shelf-life time*: how many years the data must remain secure for;
- the *migration time*: how many years it will take to upgrade the systems guarding that data;
- the *quantum threat time*: the estimated time until potential adversaries gain access to quantum computers of cryptographic significance.

The *Mosca inequality* (Figure 1) states that if the quantum threat time is less than the combined duration of the shelf-life and migration times, organizations may fail to shield their assets from quantum-enabled breaches. This document aims to offer an informed view on the proximity of the quantum threat.



Figure 1 Mosca inequality

We shed light on the quantum threat timeline by sharing and analyzing the perspectives of 26 global experts from academia and industry, involved in all aspects of quantum computing research, development, and commercialization.

Mitigating the quantum threat to cybersecurity requires a timely, well-planned transition to quantum-safe solutions.

Predicting the pace at which a *cryptographically relevant quantum computer* (CRQC) will be developed – let alone *when* it will be developed – is plagued by uncertainty, principally because building such a device requires continuously pushing the boundaries of science and engineering.

The main challenge in building a CRQC lies in the fragility of the quantum properties that a CRQC needs to harness to outperform standard computers. The building blocks of quantum computers are

quantum bits, or *qubits*, which have several possible technological implementations, all inevitably imperfect. Quantum error correction (QEC) allows one to utilize multiple imperfect *physical* qubits with fragile quantum features to encode high-quality and robust *logical* qubits.

By now, we have seen convincing experimental demonstrations that this approach works.

Nonetheless, scaling up to the numerous logical qubits needed for quantum cryptanalysis remains daunting. QEC requires a substantial resource overhead, not only in terms of the ratio of physical qubits to logical qubits, but also in terms of the continuous application of error-correcting steps.

Despite such challenges, the polled experts generally accept that a CRQC will eventually be built on the basis that no specific fundamental roadblock has been identified and that there has been steady – and at times faster than expected – progress. Quantum researchers and companies have identified and continued to achieve key milestones in their roadmaps to further scale the size and performance of current devices towards the level needed for cryptographic applications. Progress has also been fostered by the interest in the potential general usefulness of quantum computing devices in areas that range from fundamental physics to material and drug design, from optimization to machine learning. The

The progress over the last few years has led many people, both within and outside the quantum research community, to consider it quite likely that a cryptographically relevant quantum computer will be built by 2035.

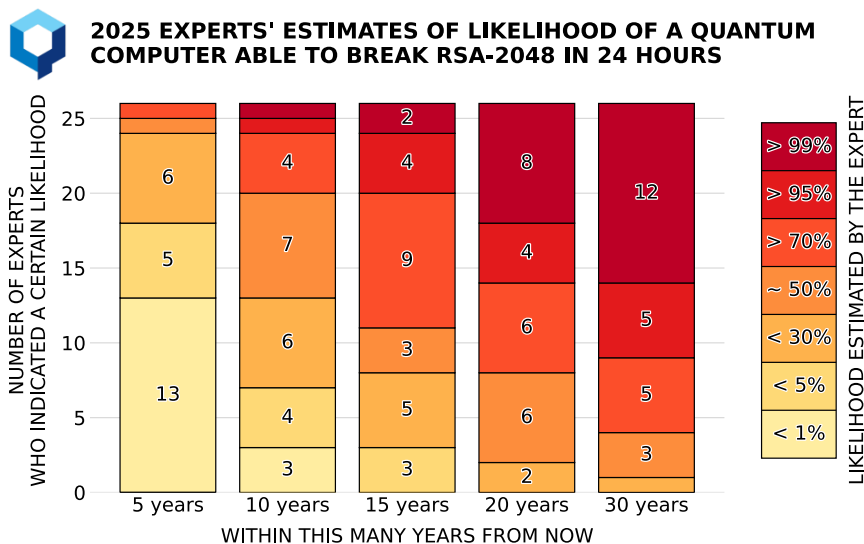


Figure 2 Experts' estimates of the likelihood of a cryptographically relevant quantum computer, aggregated by likelihood bin and timeframe.

usefulness of present and near-term quantum computers in such fields is far from well-established, but it has nonetheless generated significant interest and investments, also because it may in some cases be achieved more easily than cryptanalytic usefulness.

Like in our previous surveys, our respondents have shared their opinions on several of the topics mentioned above. Most importantly, they have provided estimates for the likelihood that a CRQC may become available within various future timeframes (Figure 2). This year, a substantial majority (19/26) felt it was more than 5%-likely already within a 10-year timeframe, with half of the respondents (13/26) indicating a likelihood of about 50% or more. This signals a substantial shift in the general belief that a CRQC may be built by the mid-2030s, compared with recent past surveys.

Another way to see this is to consider an average, opinion-based estimate of the likelihood of creating a CRQC across the overall pool of respondents (Figure 3). An “optimistic” interpretation of the responses – focused on the upper end of the likelihood ranges the respondents could choose from – leads, on average, to a ~49% estimate of a CRQC being developed within a decade and ~15% within 5 years. Even a “pessimistic” interpretation focused on the lower end of the likelihood range yields an average likelihood estimate of ~28% for a disruptive quantum threat over the next 10 years. If we compare these average likelihood estimates with those from recent past surveys (Figure 4), one cannot help but notice an upward shift, particularly for the 10-year timeframe. We stress that this means that, depending on a company's or institution's risk tolerance and needs, many organizations may unknowingly be exposed to an intolerable level of risk that demands immediate action.

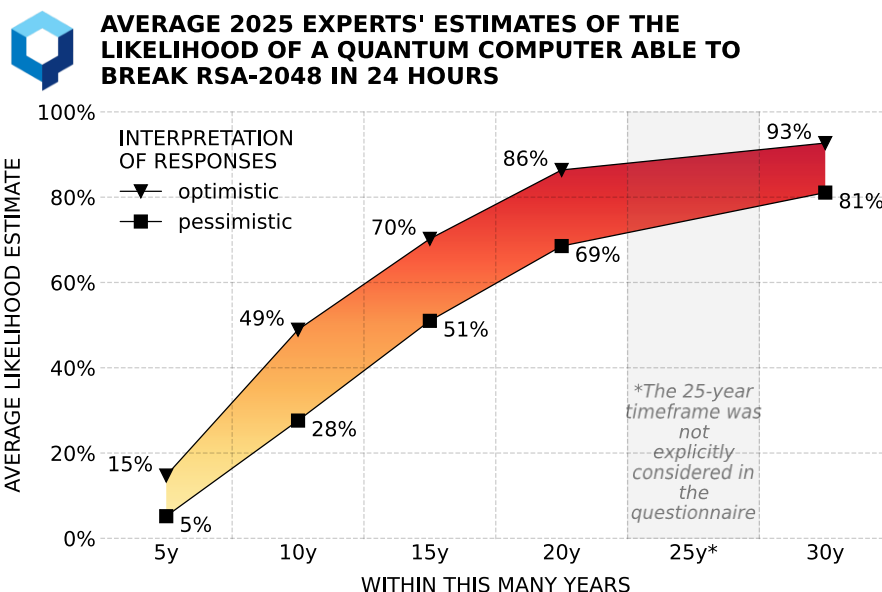


Figure 3 Averaged experts' likelihood estimates by timeframe.

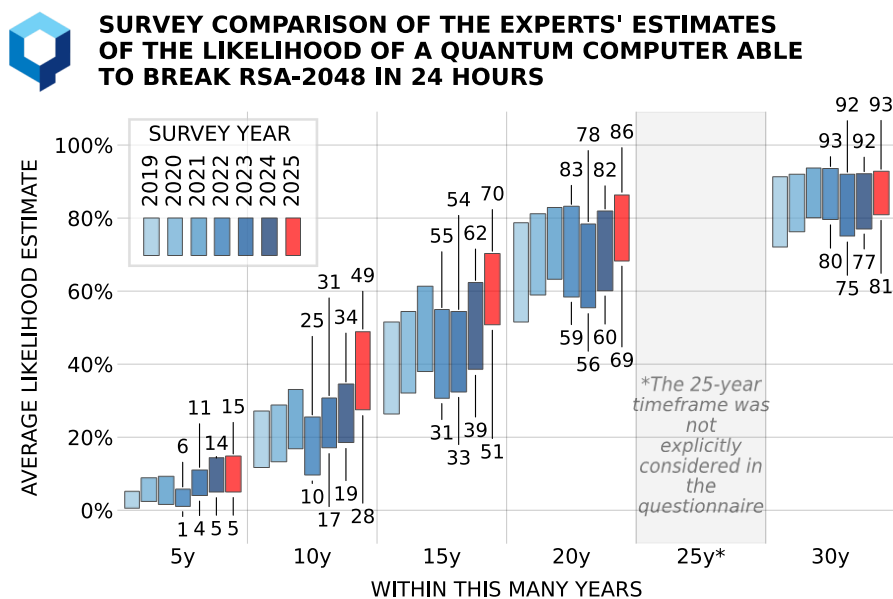


Figure 4 Comparison with previous surveys of the ranges between averaged optimistic and pessimistic interpretations of the choice of the likelihood bin, by timeframe. We explicitly indicate the values of the optimistic and pessimistic averages for the 2022-2025 surveys.

In assessing quantum risk, one should also consider that ‘covert research’ – such as that conducted by government-supported laboratories, by companies operating in stealth mode, or by malicious private entities – may influence the quantum threat timeline in significant ways. Overall, the experts agree that

this is a concrete possibility, suggesting that the quantum threat could be closer than what can be inferred from openly available information like scientific publications or press releases. Furthermore, one should consider that there can be significant and unexpected accelerations in quantum computing research. The experts identified quantum error correction as well as hardware development as the subfields of research more likely to produce such accelerations.

Many organizations may be unaware that they are currently exposed to an intolerable level of risk that requires urgent action.

The cyber risk posed by a quantum computer doesn't simply lie in exposed secrets. Suppose the quantum threat arrives before quantum-safe migrations of systems are completed. In that case, integrity, availability, and operational control across critical systems are at stake—that is, a systemic security and operational failure, not just a confidentiality breach. On the other hand, it is crucial to understand that adversaries do not need to remain inactive while waiting for the availability of a CRQC: they can already intercept, duplicate, and archive encrypted communications for future decryption—a strategy known as “*Harvest Now, Decrypt Later (HNDL)*.” This is accounted for in the Mosca inequality, which considers the required shelf life of the data.

Governments are taking the quantum threat seriously by mandating a transition to *Post-Quantum Cryptography* (PQC), for example, via the adoption of PQC algorithms and standards recently issued by the US National Institute of Standards and Technology (NIST). In Canada, a detailed roadmap for federal IT systems was released in 2025: every government department must draft a PQC migration plan by April 2026, with high-priority systems to switch to PQC already by the end of 2031. Along similar lines, the European Commission has published a Coordinated PQC Transition Roadmap that requires all EU member states to begin deploying post-quantum encryption by 2026. Notably, EU critical infrastructure operators must complete their transition by the end of 2030.

The relative positions of global competitors in the so-called “quantum race” provide another perspective on the Quantum Threat Timeline that may interest cybersecurity professionals. Most surveyed experts identify North America as the current leader in quantum computing. Many believe it will likely maintain this leadership for the next five years, but they also see China as a potential leader by then. Meanwhile, Europe is viewed as somewhat falling behind.

Overall, given the rapid progress in quantum computing, the insights from our surveys, the momentum from substantial investments in the field, and the threat from HNDL attacks, we cannot but conclude that cybersecurity leaders and managers should act promptly to ensure cyber resilience in the face of the looming quantum threat. They might do so by actively pursuing cryptographic modernization, accelerating plans for a secure transition to PQC, and developing layered defenses against current and future threats to public-key cryptography. Even organizations not yet officially required to adopt PQC can benefit from a proactive stance, reducing the risk of rushing to quantum-safe cryptographic systems later.

From the threat timeline to the migration timeline

All organizations should evaluate their urgency in proceeding with their migration to quantum-safe systems (World Economic Forum 2023; Canadian Forum for Digital Infrastructure Resilience 2024). The Global Risk Institute and evolutionQ Inc. have made available an updated Quantum Risk Assessment (QRA) methodology (Mosca and Mulholland 2025) on which such a process may be based.

Such a revised QRA methodology maintains quantum risk as the central focus while emphasizing the assessment of the organization’s current preparedness for cryptographic modernization. The latter is now being driven by mandates from cryptologic authorities and emerging regulatory guidance—including migrating to quantum-safe cryptography and strengthening the agility needed for future transitions. The methodology also explores how cryptographic risks impact organizational resilience and how dependencies might cause a failure of a key cryptographic algorithm to propagate through the infrastructure. Armed with this insight, organizations can begin planning their quantum-safe migration roadmap and enhancing their cryptographic resilience.

The need to secure information for a long time, together with the possibility of “Harvest Now, Decrypt Later” attacks, means that the transition to quantum-safe solutions must occur much earlier than the actual creation of a cryptographically relevant quantum computer.

1 Introduction

This is the 7th report in an annual series that began in 2019, intended to shed light on when a quantum computer powerful enough to be cryptographically relevant might be developed. The reports primarily rely on a recurring survey of leading experts in quantum computing research. These experts have been asked questions to evaluate the progress and pace of development in the field, and, most importantly, to offer their best estimates of when such a *cryptographically relevant quantum computer* (CRQC) may become available.

This introduction provides a brief background to help the reader understand how quantum computers threaten cybersecurity and the significant scientific and technological challenges involved in developing such computers. Some more technical information is offered via `in-depth` text boxes, and further details can be found in the Appendix, which also contains a Glossary.

1.1 Quantum computing

Quantum mechanics is our most successful scientific framework for understanding the behavior of systems such as atoms and fundamental particles, including electrons and photons.

Understanding and controlling this behavior has been crucial for the development of many modern technologies, including lasers and computer microchips, during the 20th century. In recent decades, researchers have realized that quantum features can be further exploited at new qualitative and quantitative levels, redefining the limits of what is technologically possible. With these efforts underway and already producing remarkable results, the United Nations designated 2025 as "The International Year of Quantum Science and Technology." Fittingly, the 2025 Nobel Prize in Physics was awarded "for the discovery of macroscopic quantum mechanical tunneling and energy quantization in an electric circuit," with the additional highlight that such a discovery "has provided opportunities for developing the next generation of quantum technology, including quantum cryptography, quantum computers, and quantum sensors."

Quantum computers leverage unique quantum phenomena to process information in a manner radically different from that of classical computers.

The primary obstacle in advancing quantum computing is the unprecedented requirement to maintain and control quantum behaviour at a level that has never been attempted before in human history.

While we all benefit from the technological advancements that quantum mechanics has enabled and continues to enable, quantum theory describes the world in terms that may seem counterintuitive or even absurd to the layperson. For example, a quantum system may exist in a so-called *superposition* of states that would be mutually exclusive in classical physics. One likely reason why quantum phenomena tend to evade classical intuition is that quantum effects are not apparent in everyday experience. This occurs because of the physical scales involved and, most importantly, because quantum phenomena are highly sensitive and easily disrupted by environmental interactions, which can degrade or even effectively eliminate quantum properties through a process known as *decoherence*.

In classical computing, information is stored in bits representing binary values, either "0" or "1." Quantum computing seeks to leverage quantum phenomena such as superposition to allow more sophisticated information processing and to aid the simulation and understanding of quantum systems. The basic unit of quantum information in quantum computing is the *qubit*. Unlike a traditional bit that holds a 0 or 1, a qubit can be in a superposition of both values, allowing them to "coexist" and, roughly speaking, be processed at the same time.

The monumental challenge in quantum computing research and development is to mitigate and counteract decoherence, protect fragile quantum features, and control quantum systems with high precision to execute quantum algorithms. *Quantum error correction* (QEC) is crucial for encoding quantum information into more robust logical qubits distributed across multiple inherently imperfect physical qubits, enabling reliable information processing.

A key milestone on the path toward a quantum computer has been demonstrating that QEC schemes allow surpassing the so-called "break-even" condition, meaning that an encoded logical qubit performs better than the physical qubits it is 'made from'. By using enough physical qubits per logical qubit, it is in principle possible to reduce logical errors to an arbitrarily low level. One major hurdle is that QEC requires significant overhead in terms of physical qubits per logical qubit encoded and in the complexity of physically implementing continuous error correction. This makes it daunting to scale the current quantum computing devices to quantum computers with many logical qubits.

No fundamental barrier to creating a large quantum computer is known, and such an achievement should be considered a matter of "when" and "how" rather than of "if".

Nonetheless, no fundamental barrier to creating a large quantum computer is known, and such an achievement should be considered a matter of "when" and "how" rather than of "if". Once realized, quantum computers will not only fulfill Richard Feynman's vision of efficiently simulating quantum systems (Feynman 1982) but also, by cleverly leveraging quantum features like superposition via specialized algorithms, solve a range of mathematical, optimization, and search problems at speeds unattainable by classical computers (Nielsen and Chuang 2000).

IN-DEPTH: IMPLEMENTING QUANTUM COMPUTERS

Several proposals and approaches are currently being developed to build a quantum computer. They vary in the physical substrate used—such as superconducting circuits, trapped ions, or quantum optics—and in the specific degrees of freedom that define qubits—for example, atomic energy levels or so-called quantum spin. They also differ in the techniques used to implement quantum error correction (QEC), aiming to achieve so-called *fault tolerance* (Nielsen and Chuang 2000).

1.2 Quantum threat to cybersecurity

The emergence of quantum computing introduces the risk of severe failures in current cyber-systems, both through direct attacks using quantum cryptanalysis and by undermining trust in these systems.

The looming threat posed by a Cryptographically Relevant Quantum Computer (CRQC) can be addressed through the implementation of quantum-resistant cryptographic techniques, which may be either classical or quantum-based. The former includes cryptographic algorithms based on problems that are

considered difficult even for quantum computers. In 2024, the U.S. National Institute of Standards and Technology (NIST) issued the first standards for Post-Quantum Cryptography (PQC) (NIST 2024). Additionally, tools like quantum key distribution (QKD) leverage quantum principles to establish shared secret keys for secure communication (Nielsen and Chuang, 2000). Transitioning to this new breed of quantum-safe cryptography is a complex and delicate process (Mosca 2013): it requires the development and deployment of hardware and software solutions, the establishment and adoption of standards, the migration of legacy systems, and more.

With the necessity to devote enough time to an orderly and safe transition to a ‘post-quantum world’, the urgency for any organization to complete the transition to quantum-safe cryptography for a particular cyber-system can be determined by considering three simple but important parameters¹ (Mosca 2013):

- $T_{\text{SHELF-LIFE}}$ (shelf-life time):** the number of years the information should be protected for;
- $T_{\text{MIGRATION}}$ (migration time):** the number of years needed to safely and adequately migrate the cyber-system to a quantum-safe solution;
- T_{THREAT} (threat time):** the number of years before the relevant threat actors will be able to break the quantum-vulnerable systems, an event often referred to as “Q-Day”.

If $T_{\text{SHELF-LIFE}} + T_{\text{MIGRATION}} > T_{\text{THREAT}}$, that is, if the time required to migrate the system *plus* the time for which the information needs to be protected goes *beyond* the time when the quantum threat will become concrete, then an organization may not be able to protect its assets for the required $T_{\text{SHELF-LIFE}}$ years against the quantum threat (see also Figure 1). This is the content of the *Mosca Inequality* (Mosca 2013), and is due to the possibility of “*Harvest Now, Decrypt Later*” attacks (Figure 5), which may affect current data/messages even if a CRQC is not yet available. Nonetheless, it is important to stress that the risk is not simply exposed secrets. Suppose the quantum threat arrives before quantum-safe migrations of systems are completed. In that case, integrity, availability, and operational control across critical systems are at stake – that is, a systemic security and operational failure, not just a confidentiality breach.

IN-DEPTH: QUANTUM CRYPTANALYSIS

Public-key cryptography like RSA (Rivest et al. 1978) could be broken using Shor's quantum algorithm (Shor, 1994), which provides an exponential speed-up over standard computer cryptanalysis. On the other hand, Grover's algorithm (Grover, 1996) enables a quantum computer to search a space of 2^n possible values in approximately $2^{n/2}$ steps, thereby reducing the security of symmetric-key cryptography, albeit not exponentially.

It is crucial to recognize that a rushed transition to post-quantum security systems may lead to the introduction of new vulnerabilities, which could be susceptible to conventional hacking techniques. These weaknesses might stem from design flaws, implementation errors, or overlooked details.

¹ Often, these parameters have respectively been called x , y , z in literature. Here we adopt a more explicit notation.

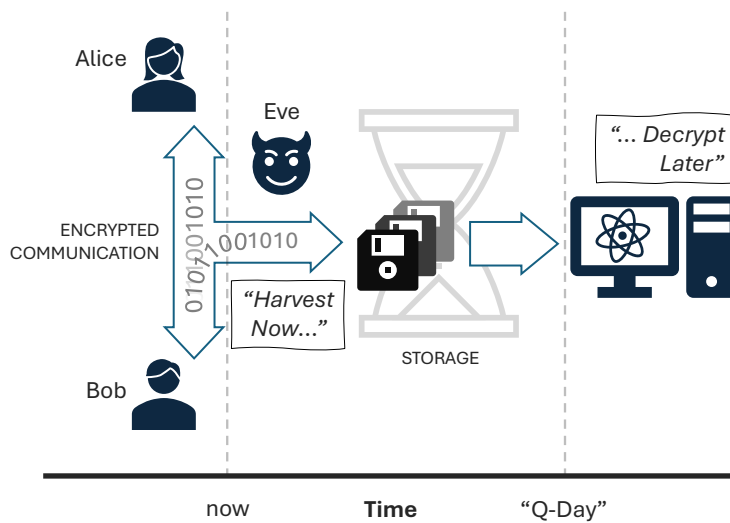


Figure 5 – Harvest Now, Decrypt Later. A malicious adversary, Eve, may already be intercepting, accessing, copying, and storing encrypted data exchanged by users Alice and Bob, delaying decryption until a cryptographically relevant quantum computer becomes available to threat actors, a future time often called “Q-Day.”

Furthermore, challenges related to interoperability and backward compatibility could emerge, complicating the migration process.

The security shelf-life time $T_{\text{SHELF-LIFE}}$ is generally organization-specific: it may be determined by business decision or dictated by regulations. On the other hand, assessing the threat timeline T_{THREAT} is challenging.

IN-DEPTH: MAXIMUM AVAILABLE MIGRATION TIME

Organizations need to assess $T_{\text{SHELF-LIFE}}$ and T_{THREAT} . The difference

$$(T_{\text{MIGRATION}})^{\text{MAX}} := T_{\text{THREAT}} - T_{\text{SHELF-LIFE}}$$

is the **maximum available migration time**, that is, the maximum time organizations have at their disposal to realize the transition safely.

There are many scientific and engineering obstacles to developing a quantum computer powerful enough to crack existing cryptographic systems. A quantum computer will be cryptographically relevant when it can run quantum algorithms for impactful cryptanalysis. Roughly speaking², this will be achieved when the number of available physical qubits—imperfect but of quality high enough—meets the number of physical qubits required to run the quantum algorithms while dealing with the significant overhead coming from logical encoding (see Figure 6). The gap between these two numbers—*available* and *required* physical qubits—has been shrinking exponentially over the past two decades, thanks to both experimental and algorithmic progress, as further confirmed by recent advances in quantum computing research (Gidney 2025).

Investments in quantum computing and associated technologies are a crucial factor affecting the pace of development, potentially shortening the timeframe for transitioning to quantum-safe systems. In recent years, funding for quantum computing research has poured in from diverse sources, including

² In general, for example, one can leverage complex trade-offs between the number of physical qubits at disposal and the length/duration of the quantum computation; the quality of the qubits and of the operations on them is also fundamentally relevant.

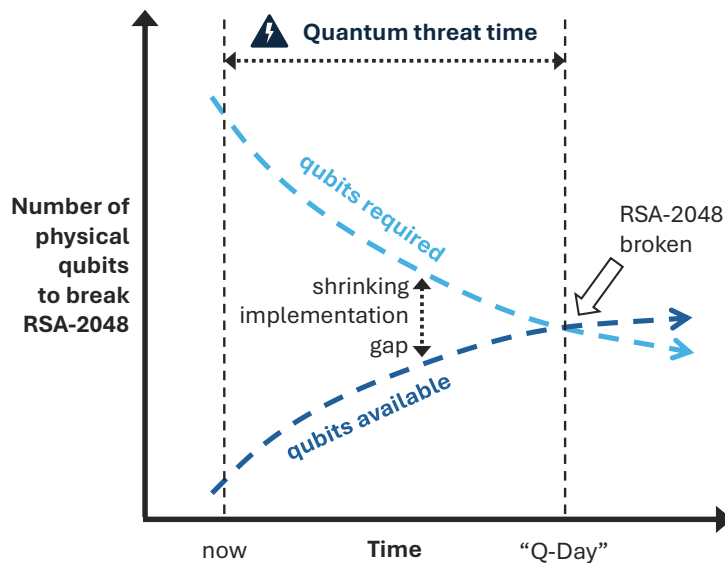


Figure 6
The Quantum Threat Time specific to RSA-2048 is the time between now and "Q-Day" for RSA-2048. The latter is when the combination of (a) an increasing number of available high-quality physical qubits coming from experimental progress and of (b) a decreasing number of required physical qubits coming from algorithmic progress will make breaking RSA-2048 possible.

government bodies, established companies, and private investors backing new startups (Kung and Fancy 2021; McKinsey & Company 2025).

In this scenario, several governments and institutions are treating the quantum threat very seriously, for example, by mandating a transition to PQC. For instance, Canada recently released a detailed roadmap for federal IT systems (Canadian Centre for Cyber Security 2025). Every department must draft a PQC migration plan by April 2026, with high-priority systems to switch to PQC already by the end of 2031. Meanwhile, the European Union released "A Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography" that requires all EU member states to begin deploying post-quantum encryption by 2026 (NIS Cooperation Group 2025). Notably, EU critical infrastructure operators must complete their transition by the end of 2030.

Rushing the migration to post-quantum cryptography might create security issues that could be exploited even by attackers who rely solely on traditional methods.

1.3 Quantum computing towards fault tolerance

It is currently believed that quantum cryptanalysis requires running quantum algorithms on a sufficiently large quantum computer that employs quantum error correction. In general, such a fault-tolerant quantum computer is needed for quantum algorithms developed to be run on an idealized error-free quantum computer. It will still take time to reach such a stage.

Meanwhile, there is undeniable excitement about the practical and commercial potential of "early-stage" quantum computers, even if not yet advanced enough to run arbitrary quantum algorithms and threaten cybersecurity. Such early computers may not adopt quantum error correction, but rather techniques for the more limited—but less demanding in terms of hardware—*quantum error mitigation*. The latter aims at managing errors and decoherence via clever circuit designs, repeated runs of the quantum computation, and classical processing of data (Eisert and Preskill 2025).

For those primarily wary of the cybersecurity risks posed by quantum computers, the interest in early quantum applications may be considered indirect. Such applications would:

- offer tangible signs and early alerts of the impending quantum challenges to cybersecurity;
- increase the likelihood of consistent funding and resources for quantum computing research, supporting the longer-term development of an error-corrected digital quantum computer with cryptographic significance.

Monitoring the development of quantum computers that are not yet cryptographically relevant, including claims and proofs of their usefulness, helps to assess both the progress in quantum computing research and the chances of continued, significant investments in the field.

1.3.1 From NISQ to FASQ

In the past years, there has been interest in so-called *Noisy Intermediate-Scale Quantum* (NISQ) devices (Preskill 2018), with tens to hundreds of physical qubits. There have been demonstrations – often controversial ones – of how such devices may go beyond what standard computers are capable of, the most famous example being perhaps the realization of so-called “quantum supremacy” (Arute et al. 2019). The controversies surrounding these demonstrations usually deal with two aspects:

- the problems/tasks considered to prove a quantum advantage may be specifically designed with that goal in mind, but not be of any practical utility;
- the claim of surpassing the capabilities of standard computers may lack the solidity of a mathematical proof and consider only the best-known classical approach/algorithm, while a better classical approach/algorithm may exist.

Regarding the second point, NISQ computers are now sufficiently large and accurate that it is generally accepted that they can perform some tasks that are too complex for even the most powerful standard supercomputers to emulate. However, practical and economically viable quantum computations have yet to be realized. While this might be possible with NISQ technology supplemented by error mitigation techniques, NISQ devices will eventually be replaced by what may be called *Fault-Tolerant Application-Scale Quantum* (FASQ) computers (Preskill 2025; Eisert and Preskill 2025), which will utilize quantum error-correcting codes. This will allow them to execute a broader range of tasks and applications. With respect to the scientific and technological gap between NISQ and FASQ computers (see Figure 7), it is crucial to recognize that:

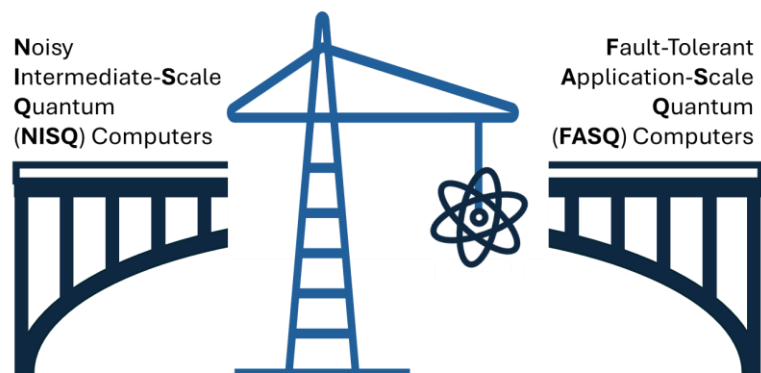


Figure 7 Quantum companies and quantum researchers are hard at work to bridge the gap between NISQ and FASQ technology, as the computational power and the utility of NISQ computers are limited by their limited size and by their relatively high error rates. In general, the extent of the NISQ/FASQ gap depends on the FASQ task/application.

- the gap ‘size’ depends on the target FASQ task/application, with the ability to run quantum cryptanalysis corresponding to a larger gap compared to other conceivably useful tasks;
- any gap will be generally overcome via progress in all aspects of quantum computing, like hardware, algorithms, and error correction schemes.

In assessing progress towards fault-tolerant computation, one can consider that logical computation may be implemented/demonstrated at different levels of capability and functionality. Barmes et al. (Barmes et al. 2025) suggested a three-level classification framework:

- Level 1:** Error correction of quantum memory, to preserve logically encoded quantum information without any processing;
- Level 2:** Limited fault-tolerant logical operations that can be implemented relatively easily on the encoded quantum information, for example, by acting in parallel on the underlying physical qubits;
- Level 3:** Full set of fault-tolerant logical operations, requiring a significant overhead in terms of physical qubits and complex active error-correction, but allowing one to run arbitrary (small-scale) quantum algorithms at the logical level.

When evaluating experimental advances in error correction and fault-tolerant computation, particularly those advertised beyond the academic setting, Barmes et al. suggest considering the demonstrated level to better put the announced results into perspective.

Finally, it is important to highlight that the U.S. Defense Advanced Research Projects Agency (DARPA) is running the Quantum Benchmarking Initiative (QBI) (DARPA 2025), which is designed to rigorously verify and validate if any quantum computing approach can reach utility-scale operation by 2033, in the sense that its computational value surpasses its cost.

KEY POINTS

- Quantum computing uses quantum mechanics principles to conduct specific calculations more efficiently than traditional computers. The basic unit of information in a quantum computer is called quantum bit, or *qubit*.
- A Cryptographically Relevant Quantum Computer (CRQC) is a quantum computer capable of breaking many common cryptographic systems once it is built, representing a major threat to data security.
- Building a CRQC requires implementing quantum error correction techniques to manage and stabilize the inherently fragile quantum states that form logical qubits.
- Developing a CRQC presents an enormous scientific and engineering challenge, as it requires both scaling the number of physical qubits and maintaining high-quality control over them, making any estimates of a CRQC timeline inherently uncertain.
- To reduce the risks from a CRQC, it is essential to switch to quantum-safe cryptographic tools. This process is complex and needs thorough planning to avoid creating new vulnerabilities.
- The urgency of transitioning to quantum-safe cryptography depends on three parameters: the projected timeline for a CRQC, the time required to migrate to quantum-safe cryptography, and the time the data needs protection for, also called shelf-life.
- If the time needed for migration, combined with the data's shelf-life, exceeds the estimated timeline for the quantum threat, assets could become vulnerable. The so-called Mosca Inequality captures this idea and accounts for the “Harvest Now, Decrypt Later” attacks.
- There is excitement about the application and commercial potential of “early-stage” quantum computers, which are not yet advanced enough to pose a cybersecurity threat. These “early” quantum computing applications could facilitate the development of a full-fledged CRQC.

2 Scope of this report

This document presents the findings of a survey conducted by evolutionQ Inc., involving 26 internationally recognized experts in quantum computing research and commercialization. Continuing the tradition of similar surveys from the past six years, the experts were invited to complete an online questionnaire about the current state of the field.

When designing the questionnaire, we emphasized clarity and specificity, especially when evaluating quantum computers as a potential cybersecurity threat. Consequently, the main question directly addresses breaking RSA-2048, whose security depends on the difficulty of factoring a 2048-bit number. The main results focus on providing opinion-based likelihood estimates of when RSA-2048 might be broken by a cryptographically relevant quantum computer (CRQC).

We aim to capture a snapshot of the experts' opinions and identify potential trends in how they change over time. Such trends can be appreciated by comparing the results of the multiple surveys that we have run year after year. In general, shifts in opinions—both at the level of individual participants who have consistently taken part in our surveys and at the level of the overall pool of respondents—may stem from steady advancements in quantum computing research, unexpected developments or challenges, and external factors like funding and investment levels. We try to capture the significance of all these influences with additional targeted questions.

Notably, we also report on the milestones the experts consider relevant to monitor in the near-to-medium term.

2.1 Related literature

Our approach centers on providing likelihood estimates based on expert opinions. Additional information and resources are available that are directly or indirectly related to estimating the timeline for developing a fault-tolerant quantum computer that could threaten cybersecurity.

For instance, Sevilla and Riedel (Sevilla and Riedel 2020) attempt to predict future progress in quantum computing by extrapolating past advancements in the field. They analyze relevant metrics—for example, roughly speaking, the number of effective logical qubits available for computation. Sevilla and Riedel focus on superconducting implementations and, like our approach, on breaking RSA-2048. The authors describe their estimates for when superconducting quantum computers might achieve this as “one piece of relevant evidence that can supplement expert opinion” and “more pessimistic but broadly comparable to those produced through the survey of experts” in the first of our series of reports. Their paper is based on data up to 2020 and has not been further updated.

Firms like McKinsey & Company are monitoring the development of quantum technologies (McKinsey & Company 2025). Their analysis is based on various sources, including publicly available data on investment levels and the number of publications and patent applications, as well as expert interviews and proprietary McKinsey analyses. Their forecasts aim at assessing the growth and future size of the market for quantum technologies, including quantum computing and quantum communication. Like Sevilla and Riedel, they also extrapolate from past progress in quantum computing to arrive at a range of dates for “Q-Day”, but do so less rigorously, as estimating the quantum threat timeline is not the central goal of their report.

Germany's Federal Office for Information Security (BSI) has issued reports on the "Status of quantum computer development" (Wilhelm et al. 2025), with particular focus on the progress towards cryptanalytic relevance. In the 2025 update, the report's authors write:

Our conservative estimate is that cryptographically relevant quantum computers are likely to be available within 15 years.

They add³:

If the [quantum-Low-Density-Parity-Check-code] demonstrators that are currently being developed perform well, this could even go down to about 10 years.

Given new developments in error correction and mitigation as well as hardware, they go on to warn the reader as follows:

A lot more can move and surprise, and [disruptive results] could accelerate the development [of a CRQC] to below a decade.

Finally, key indications about the quantum threat timeline come from the roadmaps of companies working towards the realization of fault-tolerant quantum computers, such as Google (Google Quantum AI), IBM (IBM Quantum Roadmap), IonQ (IonQ Roadmap), Microsoft (Microsoft Quantum Roadmap), Quantinuum (Quantinuum Roadmap), or QuEra (QuEra Roadmap).

KEY POINTS

- This report is part of a series based on annual surveys to collect and analyze opinions of leading experts in the field of quantum computing.
- The primary goal of the report is to provide unique insight into the quantum threat timeline based on expert opinions, complementing other approaches and sources of information.

³ See Section 4.3 and the Appendix for more information about quantum Low-Density-Parity-Check error correction codes.

3 Participants

Since the inaugural survey in 2019, we have annually reached out to leading international experts to gather diverse and insightful perspectives on progress in quantum computing. Over the years, we have endeavored to maintain the original 2019 respondents to monitor shifts in their views. Additionally, we have approached other potential participants, chosen from an extensive list of worldwide preeminent experts. Those who agree to participate are requested to fill out an online survey, but a few provided input only about the likelihood estimates of a cryptographically relevant quantum computer.

Overall, the size and composition of our experts' pool have changed each year, while maintaining a core of recurring respondents. In 2025, we collected responses from 26 experts (listed in Appendix A.1). In this section, we summarize the composition of the group graphically in terms of:

- country where they work (Figure 8),
- main research activity (Figure 9),
- their organization affiliation (Figure 10).

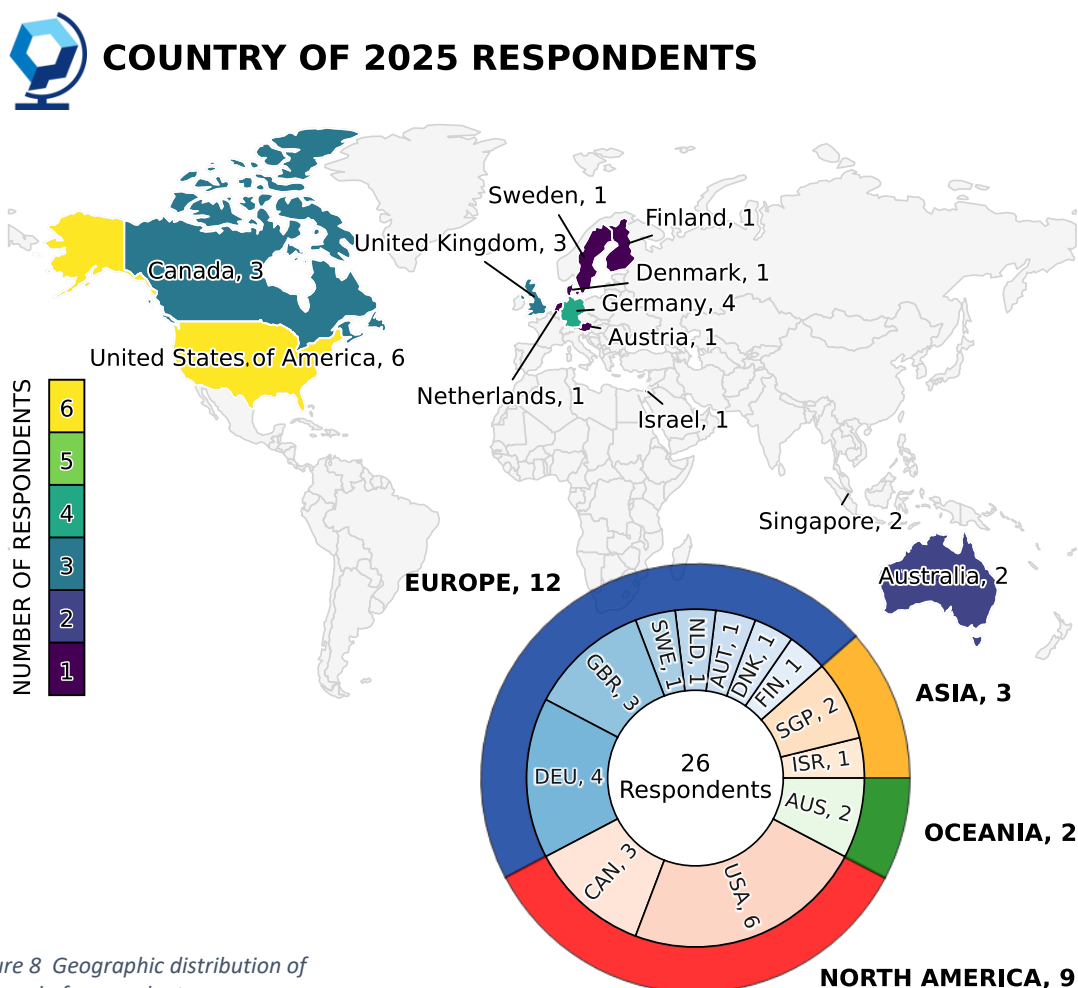


Figure 8 Geographic distribution of the pool of respondents.



RESEARCH ACTIVITY OF 2025 RESPONDENTS

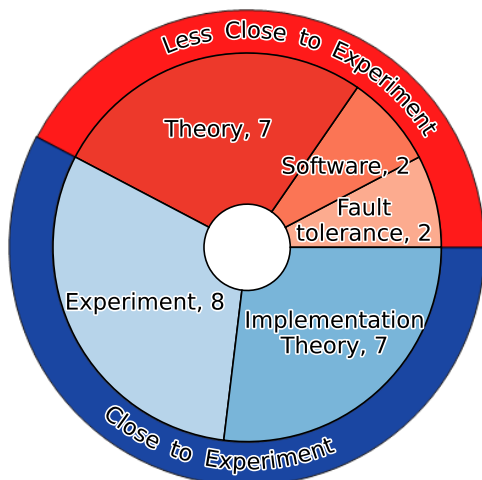


Figure 9 Research activities of the respondents.



TYPE OF INSTITUTION OF 2025 RESPONDENTS

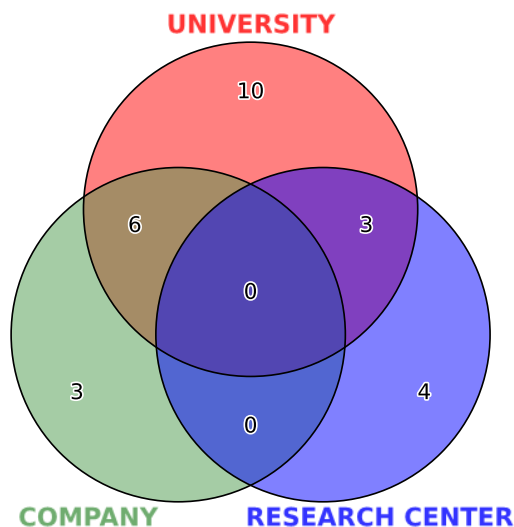


Figure 10 Typology of institutions with which the respondents are affiliated.

Our respondent group displays a diverse mix of national backgrounds, expertise, and representation from both academic and private sectors.

With respect to research activity (Figure 9), while the major division is between non-experimental and experimental research, research that is not directly experimental can vary widely. For example, implementation theory focuses on guiding, supporting, and facilitating experimental efforts. Respondents are classified under simply “theory” if their more abstract activity is not specifically related to experiments or implementations, or to fault-tolerance, or to software development.

Regarding their affiliation (Figure 10), most respondents work at universities, whereas some work at companies or research centres. Several academics have some role in—or at least collaborate closely with—external companies.

KEY POINTS

- Our respondent pool features a diverse mix of expertise, national backgrounds, and representation from both academic and private sectors, reflecting the multifaceted nature of the quantum computing community.
- The size and composition of our experts’ pool have changed each year, while maintaining a core of recurring respondents.

4 Survey results

We provide an aggregate quantitative analysis of expert responses about the following:

- the potential of various physical implementations/platforms for quantum computing (Section 4.1);
- the quantum threat timeline (Section 4.2);
- practical applications of early quantum computers (Section 4.4);
- “under the radar” research (Section 4.5);
- societal and funding factors (Section 4.6);
- potential sources of unexpected speed-up in the development of a cryptographically relevant quantum computer (Section 4.8).

We also include a selection of opinions about:

- the most promising schemes for fault-tolerance;
- key recent research developments, as highlighted by the respondents;
- near-future (that is, approximately, by mid-2026) developments that the respondents see as essential on the path to developing a fully scalable fault-tolerant quantum computer.

Respondents' comments can be quoted with their permission. Quotes may be attributed to a specific respondent or reported anonymously as "Respondent." Quotes may be lightly edited for brevity and clarity.

When appropriate, we analyze changes in responses relative to those from the previous six years. In the combined analysis of responses, we show how many respondents (or what percentage) selected a specific answer from the options given to them.

Not all the 26 respondents provided input for all questions. Also, the respondents took part in the survey during a period when new results were announced, so their input may reflect a slightly different state of the field.

Finally, some questions might have been modified or tweaked in their wording from survey to survey, but we have intentionally kept the key question about breaking RSA-2048 the same. Nonetheless, to assess how using relatively large likelihood bins may affect the accuracy of responses to such a question, we have asked respondents to provide point estimates, if willing.

Caution is advised when interpreting any trend that may emerge from a simple comparison with past responses, as it can be challenging to isolate confounding factors (see also the Appendix). However, when we observe a trend that might be potentially significant, we highlight it and, if possible and deemed appropriate, attempt to provide a rationale that might explain it.

4.1 Physical realizations

Regarding the physical implementations of quantum computers, we asked respondents to evaluate the potential of several options as candidates for fault-tolerant quantum computing. Specifically, we asked them to consider the goal of building a quantum computer with approximately 100 logical qubits within the next five years.

Over the five surveys from 2019 to 2023, we asked a similar question but with respect to a 15-year timeframe. In 2024, we shifted to a 10-year timeframe, motivated by the passage of time since the start of the survey series and by the progress in the field. This year, we further tweaked the question to focus on a 5-year timeframe. Recent advancements and the aggressive roadmaps of some hardware companies drove the change. However, most respondents noted that the new 5-year window was too short, leading some to classify systems as “not promising” within that timeframe despite their viability as a quantum computing platform on longer timescales.

“I think that 100 logical qubits in 5 years would be a stretch for any system, but it is within the realm of possibility.”

DANIEL GOTTESMAN
University of Maryland

Overall, superconducting systems seem to remain the most promising implementation (Figure 11). Nonetheless, as observed already in last year’s report, recent experimental results have heightened the interest in cold-atom platforms as leading candidates for large-scale quantum computers, putting them almost on par with superconducting systems. One reason is that cold-atom systems can scale up quickly and relatively easily, at least as the number of physical qubits goes, compared to other implementations. In turn, this, together with flexible interactions, also allows such systems to encode relatively large numbers of logical qubits. Also, while one disadvantage of cold-atom systems compared to, say, superconducting systems, is their operating frequency, speed is not quite covered as a relevant

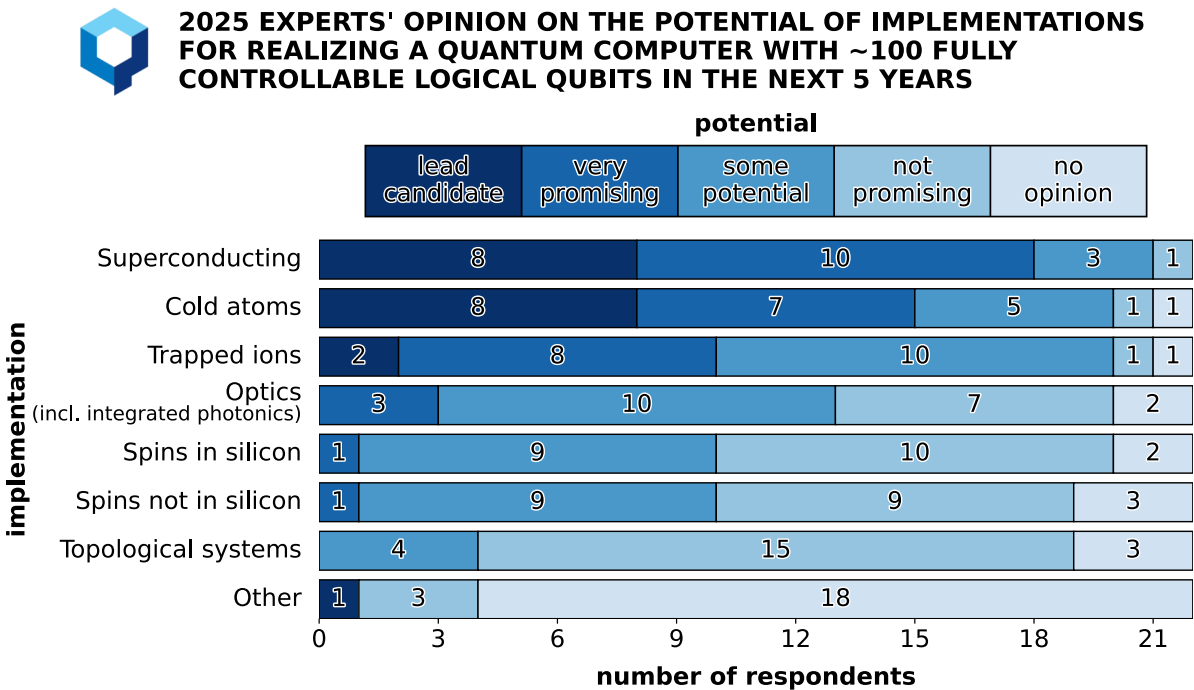


Figure 11 Experts’ opinions on the potential of various physical implementations of quantum computers.

parameter in our question. All these considerations might explain why **Bill Coish**, a professor at McGill University, commented:

Based on recent advances (algorithms performed on several 10's of logical qubits), I think it is very likely that Rydberg atoms will soon reach >100 logical qubits with an error rate below that of the physical qubits, but all other implementations face significant roadblocks to reaching this number within 5 years.

With respect to the 5-year timeframe, **Nicolas Menicucci**, a professor at RMIT University, wrote:

Recent experiments in superconducting circuits, trapped ions and neutral atoms have demonstrated small numbers of logical qubits whose lifetimes or logical operations beat those of the underlying physical qubits, but always in tightly constrained settings and at much smaller logical qubit counts. Scaling this to ~100 fully controllable logical qubits, with real-time error correction and a practical control stack, is, in my view, far beyond what can credibly be achieved in 5 years. [...] On a 5-year scale, the only serious uncertainty comes from the possibility of a completely disruptive, currently unseen innovation.

Among “Other” systems, Menicucci points to bosonic encodings⁴ as the most plausible long-term route to scalable fault tolerance, but again not on a 5-year horizon.

On the other hand, **Stephanie Simmons**, Chief Quantum Officer of Photonic Inc. and a professor at Simon Fraser University, is more optimistic and thinks that each of the platforms currently taking part in the DARPA Quantum Benchmarks Initiative program (see Section 1.3.1) could plausibly attain ~100 logical qubits within 5 years, in particular, any of those that will be selected for Stage C of the program.

Klaus Mølmer, a professor at the Niels Bohr Institute of the University of Copenhagen, highlights the possibility of hybrid systems that combine the advantages of, for example, superconducting systems with those of spin/atom systems.

KEY POINTS

- Several physical implementations of quantum computers are presently being developed; they differ in the kind of physical systems and physical encodings that are utilized as qubits.
- Each implementation has advantages and disadvantages, which become even more significant when scaling to many qubits.
- When considering the implementation of logical computation with approximately one hundred logical qubits outperforming the underlying physical qubits, most respondents are skeptical that this could be achieved within any current platform in the next five years.
- This does not mean such a feat is impossible, but it might require fast progress/innovation.

⁴ Optical or microwave continuous-variable architectures utilizing the so-called bosonic codes for error correction; see also the Appendix.

4.2 Quantum factoring

In this survey, as with previous ones, the most relevant information about the quantum threat timeline comes from the experts' assessment of the likelihood of developing a quantum computer capable of breaking RSA-2048 quickly, in response to the following question (see also Appendix A.3):

Q: Please indicate how likely you estimate it is that a quantum computer able to factorize a 2048-bit number in less than 24 hours will be built within the next 5 years, 10 years, 15 years, 20 years, and 30 years.

Estimates on the practical requirements to achieve such a feat, also considering the imperfections of physical implementations, were presented, for example, in (Gidney and Ekerå 2021) and (Gheorghiu and Mosca 2025). A recent work claims that a fault-tolerant quantum computer using fewer than one million physical qubits could factor a 2048-bit RSA key in less than a week⁵ (Gidney 2025).

In our surveys, we 'only'⁶ ask to provide a likelihood estimate, expressed as choosing one of seven likelihood bins, ranging from "extremely unlikely (<1% chance)" to "extremely likely (>99%)" (see Appendix A.3 for details). It is important to emphasize that these bins are not evenly sized in terms of likelihood range: the ones mentioned cover a 1% range, while the intermediate bin, "neither likely nor unlikely (~50%)", spans a 40% range, from 30% to 70% likelihood. Using a few bins of different sizes was a design choice, specifically the result of considering:

- the significant uncertainty involved in handling this kind of estimate;
- that from a risk assessment and risk management perspective, when dealing with highly consequential risk, it matters more whether the risk increases from "<1%" to ">1% but <5%" than whether it increases from "<30%" to ">30% but <35%".

We have kept the question-and-answer options consistent across all surveys. Starting with the 2024 survey, to understand the impact of using a few relatively large, uneven likelihood bins, we asked respondents to optionally provide point likelihood estimates—i.e., to pick a single value rather than a range bin.

"The hardware community is making very good progress at producing larger and better processors with 100s-1000s of qubits on several different physical platforms. In the next few years, I believe we will possibly stay in this range in terms of the number of qubits and continue to improve the quantity of these qubits, allowing more flexible connectivity, developing better/more efficient operations, etc. [...] I think on the 10-15 year scale, we may start to see systems that are sufficiently large and robust to attempt breaking RSA."

YVONNE GAO

National University of Singapore

⁵ In our question, we speak of "24 hours" for the sake of concreteness, and the central claim of (Gidney and Ekerå 2021) was about an 8-hour computation. These differences in computation time are relatively immaterial in comparison to the 20-fold reduction in the number of required physical qubits achieved in (Gidney 2025) compared to (Gidney and Ekerå 2021).

⁶ Many participants in our annual surveys have emphasized the inherent challenges in making such predictions.

The primary findings of our yearly survey are illustrated in Figure 12, which provides the aggregate distribution of the responses of the experts⁷. It depicts the estimated increase in the likelihood of the quantum threat as we transition from the near future to the more distant future (see also Figure 13). Some highlights of the collection of likelihood estimates are summarized in Table 1.

We observe significant variation in expert opinions: some are optimistic, while others are more cautious about how quickly quantum computers will be developed. This is also illustrated in Figure 14 and Figure 15; in the latter, the individual pattern of responses for each expert is displayed.

For some respondents, their highest estimated likelihood of the quantum threat occurs before the 30-year mark. For a subset of these individuals, the peak likelihood is lower than the maximum identified in our survey. Such perspectives may be seen as an acknowledgment of potential unforeseen technological hurdles or even insurmountable barriers. One respondent explicitly explains why their estimates were capped at 70%:

In approximately 20–25 years' time, I expect that the objective will either have been reached, or that we will have identified and understood one or more key obstacles preventing it from being reached. This explains why I will not go above a 70% chance in the above estimates.

Figure 16 provides some insight into the effect of using a few relatively large likelihood bins of variable size. It plots the trajectories of the point estimates of the respondents ($n = 15$) who chose to provide those in addition to a choice of likelihood bins. The graph further stresses the wide diversity of estimates, particularly at the 10-year timeframe.

Correspondingly, the experts' comments show a range of opinions on the likelihood estimates. It is worth highlighting that even some of the more cautious respondents often emphasize the potential for sudden breakthroughs and accelerations.

In his comment, elaborating on the rationale for his likelihood estimates, **Bill Coish** provides a succinct summary of the engineering/practical challenges faced by several physical implementations of quantum computing:

Photonic platforms suffer from statistical photon loss and heralding to produce resource states, which makes scaling very difficult unless the losses can be reduced to levels that are almost unimaginable right now (requiring improvements in engineering or quality control).

Superconducting qubits require high-quality interconnects between dilution refrigerators to scale. These interconnects must be designed in a way that is tolerant to microwave photon loss – again, requiring physical elements that are just difficult to engineer to sufficient quality.

"The core bottleneck will be to connect sufficiently many units sufficiently well to arrive at a scalable fault-tolerant quantum computer. In particular for superconducting architectures, this is challenging."

JENS EISERT
Freie Universität Berlin

⁷ The same data are provided in a more data-sharing-friendly table in Appendix A.4 .

Ion-trap quantum computers are limited by their ability to distribute entanglement deterministically.

Neutral-atom (e.g., Rydberg-atom) quantum computers tend to be slow, in particular for initialization and readout.

Spins in semiconductors have long-term potential for a scalable architecture, but the engineering challenges required to reduce the effects of charge noise and crosstalk, as well as the number of control lines that must be fed into a dilution refrigerator, make fast scaling very difficult without improved materials and engineering.

He goes on to conclude the following:

There is the possibility that any one of these implementations has an engineering breakthrough that allows rapid scaling, even in the next 5 years, but I consider that to be an unlikely event (probability ~5%). Since the limitations are primarily at the level of engineering, and the incentives to produce this device are so strong, given >15-20 years, I believe it is very likely that RSA will be broken by some implementation.

“Both physical and time resources have been reduced dramatically, and demonstrations proceed apace on five or more promising architectures.”

STEPHANIE SIMMONS
Photonic Inc.
& Simon Fraser University

Nicolas Menicucci similarly provides an assessment of platforms, but focusing mainly on two physical implementations, superconducting circuits and photonics, which he describes as facing complementary issues:

Superconducting circuits run into hard scaling problems, [forcing us to consider] a modular, multi-cryostat architecture with coherent interconnects. [..]

On the photonics side, [..] integrated optics and room-temperature operation offer a more convincing route to very large device sizes and high connectivity. [T]he bottleneck is not interconnects but rather the initial preparation of good encoded qubits.

No current platform has solved all of these problems at once.

Daniel Gottesman, a professor at the University of Maryland, reminds us that there are non-technical factors that play a large role in influencing the quantum threat timeline:

There are many technical uncertainties, but another big one is funding and social commitment to quantum computing. Right now, there is a great deal of enthusiasm for supporting the development of quantum computers, but if that dries up – perhaps because progress is slower than some supporters are expecting – the timeline could be extended considerably.

Alexandre Blais, a professor at the University of Sherbrooke, stresses that some of the most sophisticated resource estimates we have, like (Gidney 2025), and that we may use to gauge how far we are from breaking RSA-2048, focus on a relatively specific (type of) platform and choice of error correction, and may lead us to underestimate the potential of other platforms and other error-correcting schemes to shorten the quantum threat timeline.



2025 EXPERTS' ESTIMATES OF LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS

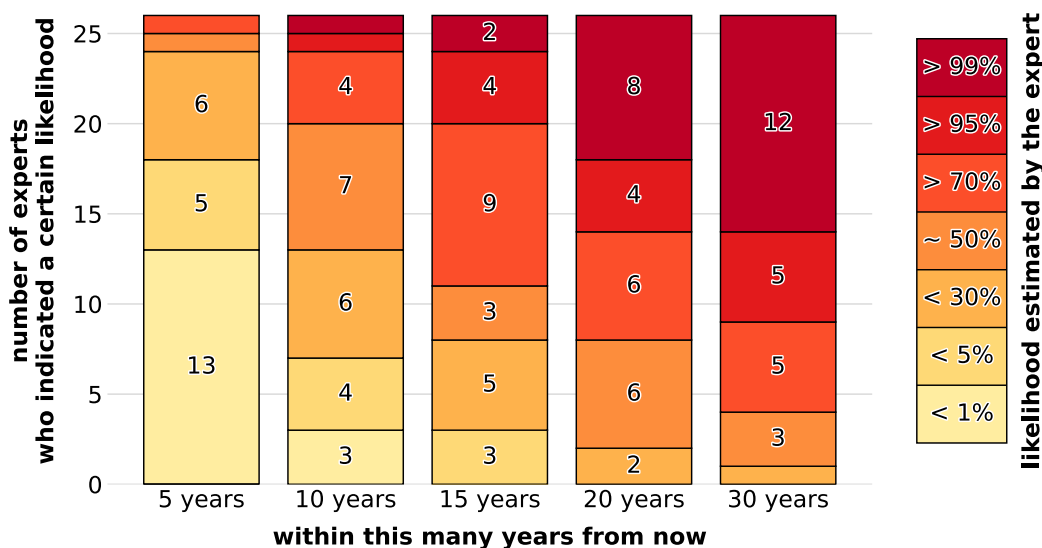


Figure 12 Experts' estimates of the likelihood of a cryptographically relevant quantum computer, aggregated by likelihood bin and timeframe.



2025 EXPERTS' ESTIMATES OF LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS - SHIFT BY TIMEFRAME

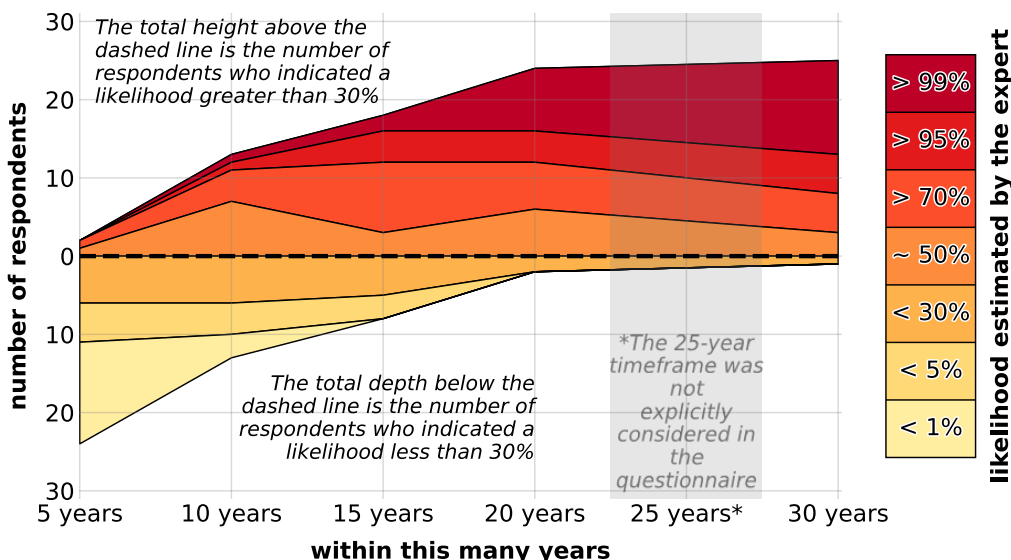


Figure 13 A stacked area chart that conveys the same data as Figure 12, offering a clearer view of how likelihood estimates shift from short-term to long-term periods: for each timeframe, the total height above the dashed line indicates the number of respondents who estimated a likelihood of at least 30%. Please note the inclusion of a dummy 25-year timeframe.

Table 1 Summary of the analysis of the experts' likelihood estimates

TIMEFRAME	WHAT ONE MAY EXPECT BASED ON THE EXPERTS' OPINIONS ON THE LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK THE RSA-2048 CRYPTOSYSTEM (IN 24 HOURS)
NEXT 5 YEARS	Half of the respondents (13/26) rate the emergence of a CRQC as $\geq 1\%$. Over 30% (8/26) see a $\geq 5\%$ chance in this horizon—including one at “~50%” and one at “likely” ($>70\%$). <i>Overall, the majority view the short-term risk as low, though a substantial fraction view this tail risk as non-negligible.</i>
NEXT 10 YEARS	A substantial majority of respondents (19/26, or 73%) estimate a likelihood larger than 5%. Also, opinions begin to spread: half of the respondents (13/26, 50%) still place the chance below 30%, but the other half judge it $\sim 50\%$ or higher, including 6/26 (23%) who say “likely” ($>70\%$) or stronger (one “very likely” and one “extremely likely”). <i>We conclude that there is a very significant chance that a CRQC will be built within this timeframe, although the experts are divided.</i>
NEXT 15 YEARS	The balance shifts: 18/26 (69%) assess the likelihood of a CRQC at $\sim 50\%$ or higher, and 15/26 (58%) call it “likely” or better ($>70\%$), including two “extremely likely.” <i>Within this timeframe, a clear majority considers a CRQC at least as likely as not, with many judging it probable.</i>
NEXT 20 YEARS	Expectations become strong: 24/26 (92%) put the probability at $\sim 50\%$ or higher; 18/26 (69%) rate it “likely” or better, and 12/26 (46%) judge it $\geq 95\%$ likely (eight “extremely likely”). <i>In this horizon, a CRQC is seen as substantially more likely than not, with many viewing it as near-certain.</i>
NEXT 25 YEARS	We did not directly probe this timeframe in our questionnaire, as we believe the unavoidable uncertainty involved in the estimates does not warrant a such a fine-grained distinction between what may happen between 20 years and 30 years from now. In some graphs, this timeframe may be included by showing interpolated values to preserve a linear timescale.
NEXT 30 YEARS	<i>Consensus approaches inevitability: 22/26 (85%) judge the outcome “likely” or better, including 17/26 (65%) at $\geq 95\%$ likely and 12/26 (46%) “extremely likely.” Only one respondent rated it “unlikely.” In general, experts expect a CRQC capable of breaking RSA-2048 to be realized within this horizon.</i>

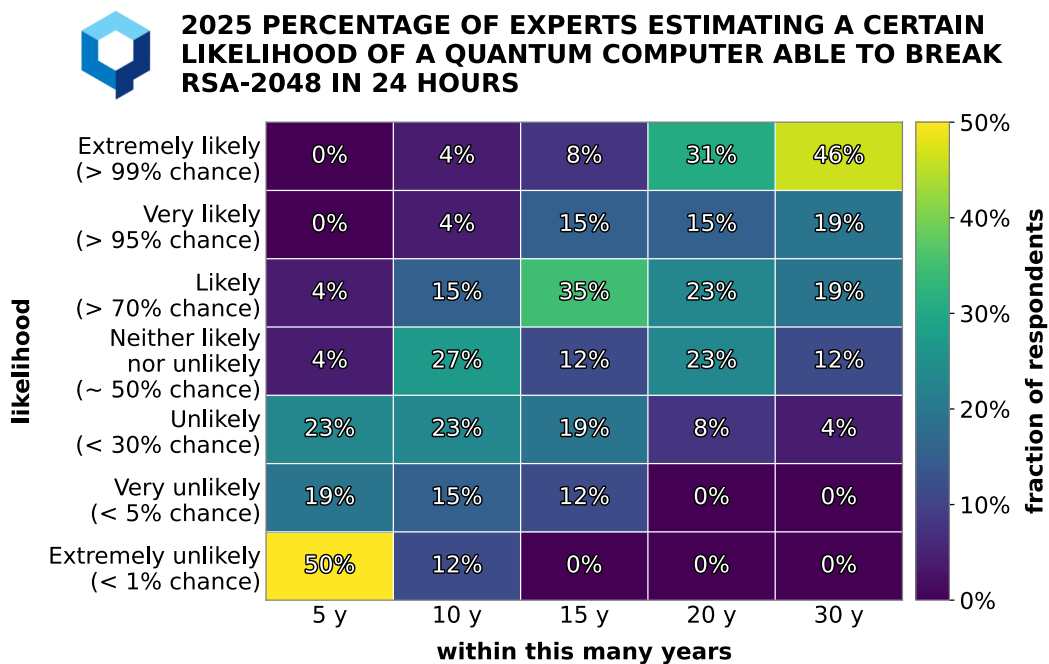


Figure 14 Heatmap of the respondents' likelihood estimates based on the choice of likelihood bins.

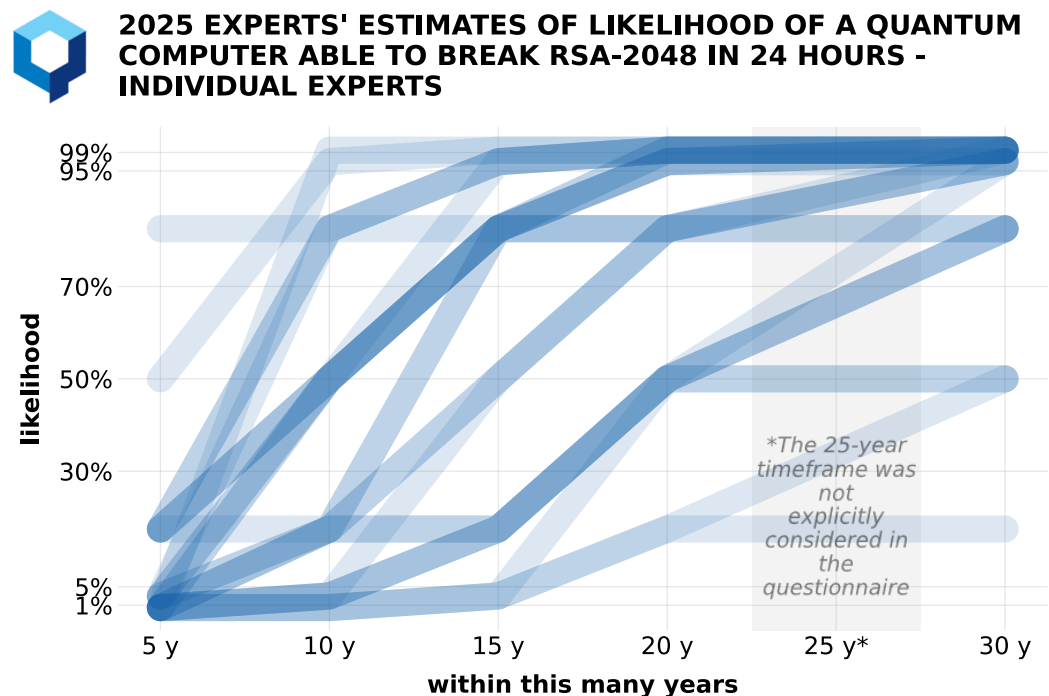


Figure 15 Likelihood estimates of the individual experts. The vertical values are the intermediate ones for the chosen likelihood bins.

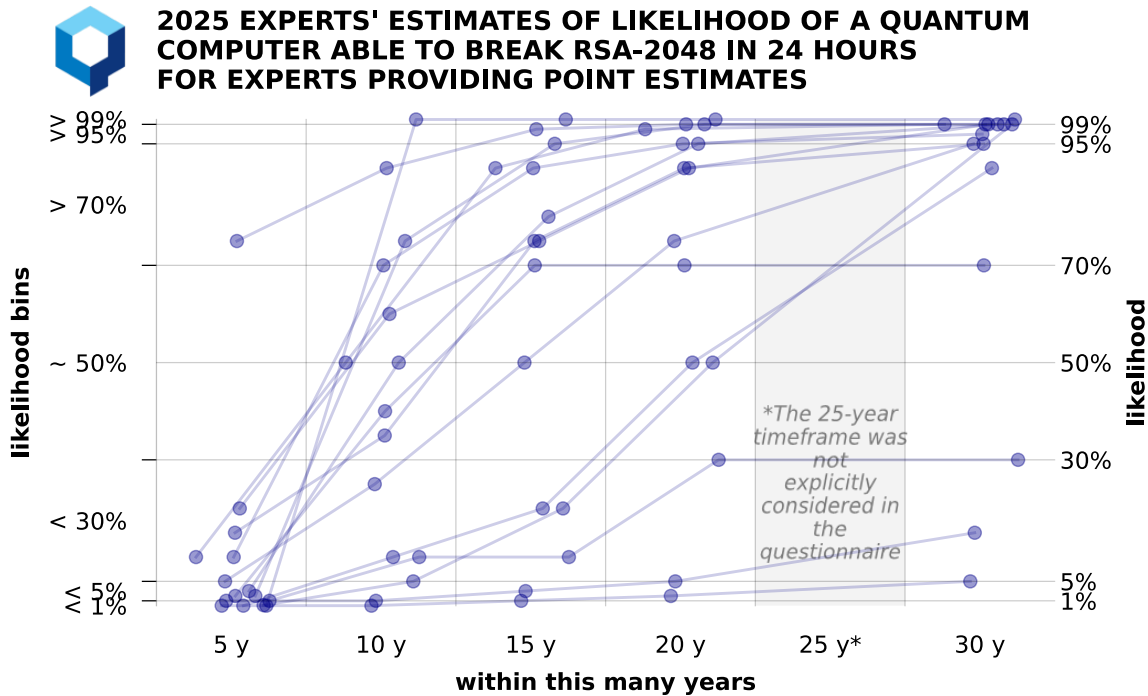


Figure 16 Individual-respondent trajectories of the likelihood point estimates, for the 15 respondents who provided them. For reference, the likelihood bin intervals used for the bin-based estimates are indicated on the left vertical axis.

4.2.1 Coarse-grained likelihood estimates

We aim to summarize the experts' insights to arrive at a single likelihood estimate. We do this by averaging the experts' estimates.

We may interpret the choice of one of the likelihood bins, e.g., “likely”, as suggesting a numerical probability within the range associated with the bin, i.e., in this case, a probability greater than 70% but less than 95%. In general, we do not know what the best point estimate of each respondent for each timeframe would have been, although we have this information for a limited set of 15 respondents (see Figure 16).

Even when interpreting the likelihood estimates “pessimistically” as the lowest compatible probability for each likelihood bin, the average probability of the disruptive quantum threat over the next 10 years is already around 28%.

We take a conservative approach and consider the two extreme alternatives where each respondent is assigned either the higher or the lower of the extreme values of the bin they picked. This can be roughly described as considering a “pessimistic interpretation” or, alternatively, an “optimistic interpretation” of the answers’ ranges. This approach allows us to calculate an average cumulative probability distribution for both interpretations. Had each respondent selected a precise estimate within the respective range, the average estimate for the likelihood would lie between the optimistic-interpretation and pessimistic-interpretation curves. In turn, the latter two curves provide some notion of uncertainty about the average probability assigned by the experts, largely reflecting the width of the likelihood bins. An intuition about the dispersion of the estimates is provided by Figure 14 and Figure 15.



AVERAGE 2025 EXPERTS' ESTIMATES OF THE LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS

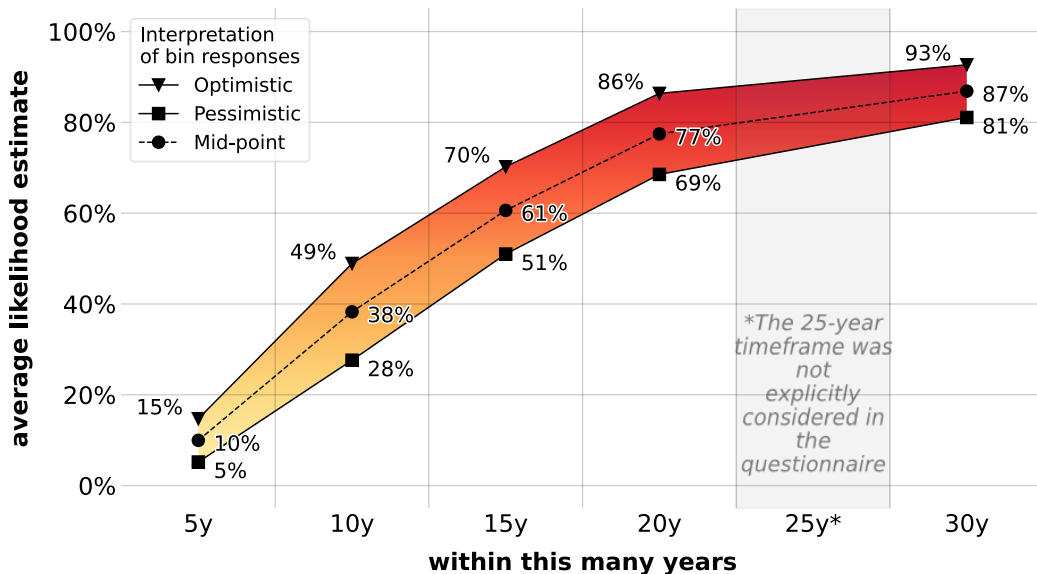


Figure 17 Averaged experts' likelihood estimates by timeframe.



2025 EXPERTS' ESTIMATES OF LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS FOR EXPERTS PROVIDING POINT ESTIMATES

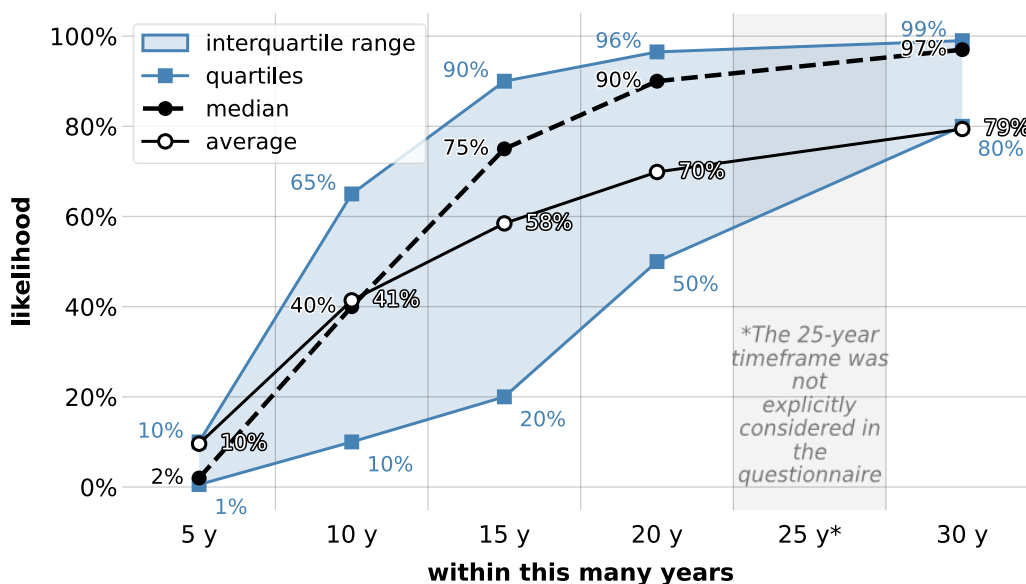


Figure 18 Average, median, and quartiles for the limited number (n = 15) of respondents who provided best point likelihood estimates.

In Figure 17, we also show a mid-point estimate, which should not be interpreted as the best estimate. More details on the method are provided in Appendix A.4 .

One can also appreciate the dispersion of the likelihood estimates by examining the distribution of the point estimates for respondents who chose to provide them (see Figure 16). Only 15 respondents volunteered this information, so it does not reflect the same diverse range of views as the entire group. Additionally, we have discussed why we didn't primarily ask respondents to provide such specific estimates. Overall, we do not present the analysis of point estimates as our main finding. Nonetheless, it is useful to consider an aggregation of those responses to compare with Figure 17: average, median, and quartiles of the point estimates are presented in Figure 18. In the Appendix, we also provide a graph that combines the bin estimates of those who did not provide point estimates with the available point estimates (Figure 33).

In general, Figure 17 should be interpreted cautiously, as it is a coarse-grained summary of our respondents' opinions, but it offers valuable insights. For example, even in a 'pessimistic' interpretation of responses, as the lowest compatible probability for a given likelihood range, the average probability of the quantum threat associated with the above-described analysis is already ~28% in the next 10 years and growing quickly in the timeframes that follow. Still within a 'pessimistic' interpretation, the average estimated probability is ~51% by the 15-year mark, and ~69% by the 20-year mark.

It is worth stressing that skewed distributions and/or outliers may affect the significance and interpretability of averages. Figures 14, 15, 16, and 18 convey insightful information about the distribution of responses. They show how responses spread out at 10 and 15 years, where the experts' estimates may vary more because there is greater uncertainty about the rate of progress. On the other hand, the response distribution is quite skewed in the shortest timeframe of 5 years – most experts think a CRQC is relatively unlikely – and in the longer timeframes of 20 years and 30 years – most experts believe a CRQC is quite likely. See also Table 1.

4.2.2 Comparison with previous years

Our series of surveys, which began in 2019, enables us to monitor changes in likelihood estimates across different surveys. We believe this is helpful for the following two reasons:

- it provides information on whether the sentiment expressed by the experts is becoming more pessimistic or more optimistic, as their opinions get affected by changing circumstances and recent progress; in turn, a change of sentiment may be interpreted in terms of a likely slowdown or speedup for the development of a quantum computer;
- it provides “differential” information that is conceivably less dependent on the baseline attitude of the pool of experts.

Comparing the results of the current survey with those of the surveys of the past few years, we find a significant increase in the overall belief that a CRQC might be built by the mid-2030s.

We emphasize that, although caution is already recommended when interpreting single-survey data, comparing year-to-year data involves additional potential pitfalls, due to factors such as changes in the composition and size of the respondent pool—particularly relevant when dealing with small respondent pools—and the relatively wide and unevenly spaced likelihood intervals we consider.

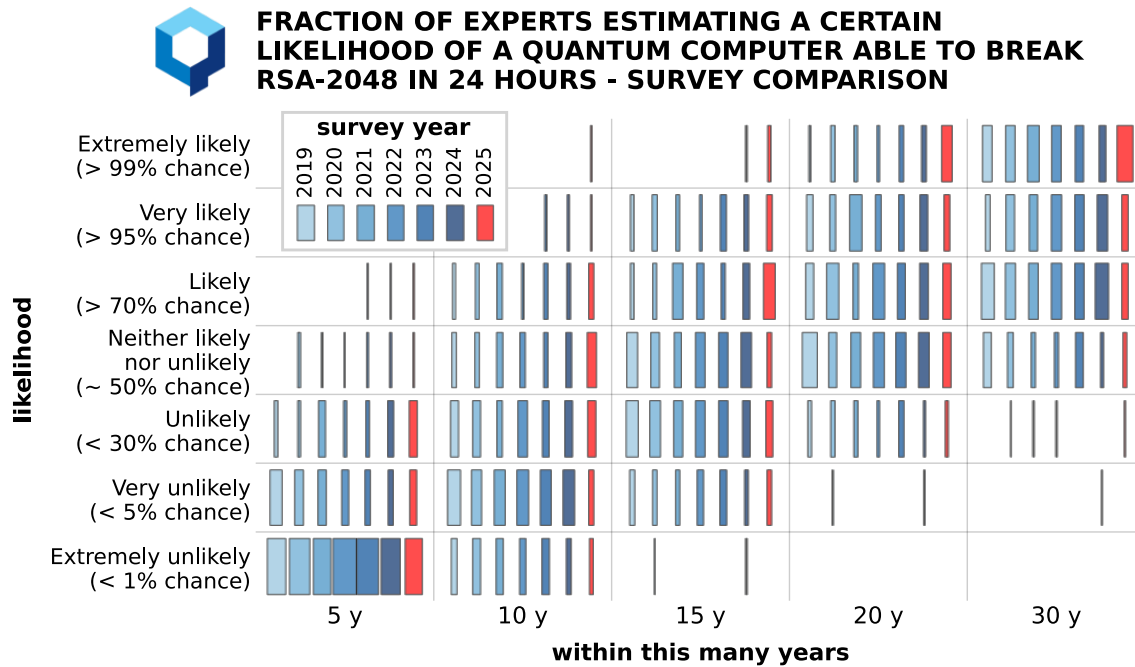


Figure 19 Fraction of respondents indicating a particular likelihood bin, by timeframe and survey year. The wider a bar, the larger the fraction. The survey year is color-coded.

In Figure 19, we plot the distribution of the likelihood estimates for each survey conducted so far – seven surveys, from 2019 to 2025. We use distributions rather than the absolute number of respondents so that we can compare surveys with different numbers of respondents. In Figure 21, we plot the average likelihood intervals for each survey, similarly to what was done in Figure 17 for just the 2025 survey.

This year, a substantial majority of respondents (19/26, or 73%) felt a CRQC was more than 5%-likely already within a 10-year timeframe, with half of the respondents (13/26) indicating a likelihood of about 50% or more. This is the highest level of optimism towards the realization of a CRQC within a 10-year horizon that we have recorded since the beginning of our series of surveys. This is reflected also in the average likelihood estimates (Figure 21).

Figure 20 and Figure 22 compare survey responses across the years by shifting the likelihood plots to match forecasts in absolute time. For example, an estimate made in 2020 for 10 years into the future (2030) should roughly be compared with an estimate made in 2025 (this year) for 5 years into the future. In Figure 20, we compare the mid-point averages for all the surveys conducted so far. In Figure 22, for the sake of readability, we only plot ranges for the 2022-2025 surveys.

A reason to focus only on the most recent surveys when fully comparing surveys in absolute time is that both the likelihood estimates and the opinions about funding dynamics (Section 4.6.1) suggest that the COVID-19 pandemic and the following tightening of venture capital acted like a watershed period, leading to a recalibration of expectations.



EXPERTS' AVERAGE RANGE ESTIMATES OF THE LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS

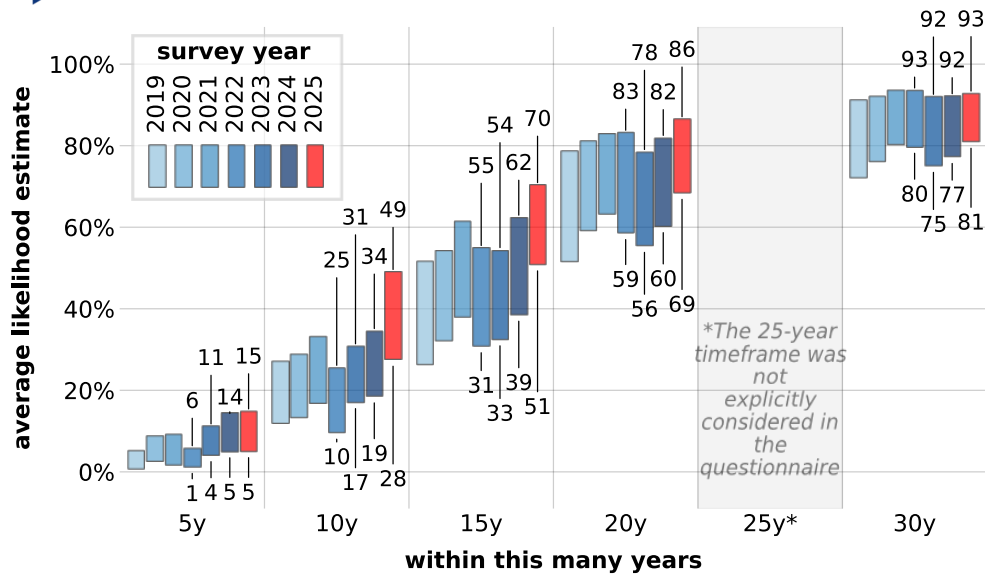


Figure 21 Evolution of the average likelihood estimates by the experts in surveys about the quantum threat timeline conducted so far. The survey years are color-coded. We explicitly indicate the values of the optimistic and pessimistic averages for each timeframe for the 2022-2025 surveys.



EXPERTS' AVERAGE MIDPOINT ESTIMATES FOR THE LIKELIHOOD OF A DIGITAL QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS, AS FUNCTION OF TIME

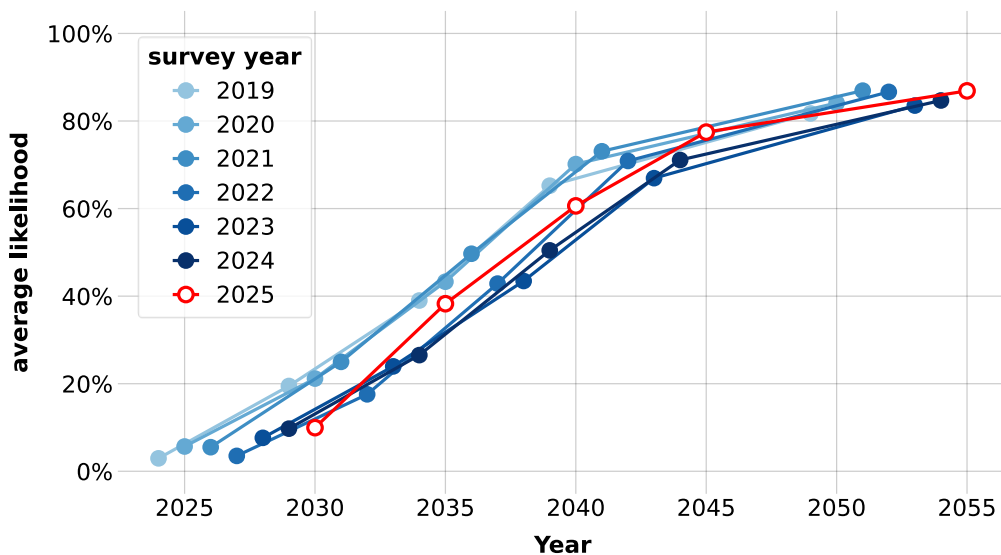


Figure 20 Comparison of the change of the coarse-grained estimates from survey to survey. The mid-point for coarse-grained estimates presented in Figure 17 for the 2025 survey is plotted for the surveys since 2019. The plots are shifted so that the estimates produced in each survey align with respect to absolute time.

Having made this observation, we stress that this year's results show a significant increase in the overall belief that a CRQC might be built by the mid-2030s, compared with surveys from the past three years. We consider this as perhaps the main takeaway from the latest survey. That the 2019-2021 surveys suggested an even higher likelihood for the same mid-2030s timeframe is of interest, but we caution against considering it as a sign of a slowdown of progress in the field in the last few years.

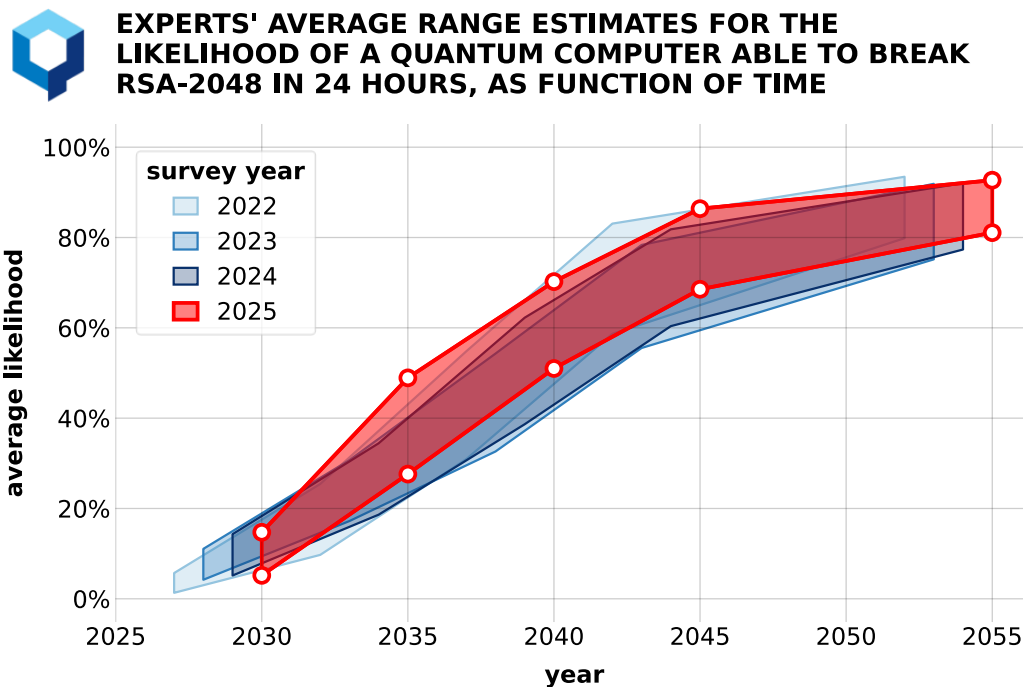


Figure 22 Comparison of the change in the coarse-grained estimates from survey to survey. The kind of range for coarse-grained estimates presented in Figure 17 for the 2025 survey is plotted also for the 2022-2024 surveys. The plots are shifted so that the estimates produced in each survey align with respect to absolute time.

KEY POINTS

- Year after year, we have asked a pool of experts in quantum computing to provide their best estimate of when a quantum computer will be able to perform a specific cryptographically relevant task: breaking RSA-2048 in 24 hours.
- The experts show a wide range of opinions about what to expect in 10 or 15 years from now, but, on average, the likelihood of a cryptographically relevant quantum computer (CRQC) increases rapidly in the relatively near future.
- Even in a 'pessimistic' interpretation of the likelihood bins that corresponds to the lower end of each bin, the average probability overall assigned to the disruptive quantum threat is already ~28% in the next 10 years.
- In an 'optimistic' interpretation corresponding to the higher end of each likelihood bin, the average likelihood is the highest in our surveys so far, for all timeframes except the 30-year one. For the next 10 years, it sits at 49%.
- This year's results show a substantial increase in the overall belief that a CRQC might be built by the mid-2030s, compared with the surveys from the past three years.

4.3 Most promising scheme for fault-tolerance

Fault-tolerance will be achieved through a combination of enhanced hardware performance and an appropriate error-correction scheme (see Appendix A.2). We have asked the experts to share their opinions on the most promising among such schemes.

A straightforward answer is not possible; in the words of one respondent:

Quantum error correction is currently a very active research area. As time progresses, it is likely that we will see more advances. This [is particularly true] as systems are scaled up. We may also see adaptations to various hardware architectures, hybrids of error-correction schemes, and so forth.

Correspondingly, the responses of the experts were relatively nuanced, highlighting, for example, that the choice of error-correction code may vary by architecture. Such a dependency arises from factors such as bias in the kinds of errors the underlying physical qubits may experience, or the connectivity, that is, whether direct interactions between the physical qubits are limited to nearest neighbours or can also be long-range. Even the same architecture may use different codes depending on the tasks – for example, memory vs computation.

Some variation of the surface code is still considered a leading choice for superconducting implementations, but quantum Low-Density Parity-Check (qLDPC) codes (see Appendix) are actively being developed and considered very promising, if not the future, by some. This is because such codes improve the encoding rate, meaning fewer physical qubits are needed per logical qubit. One price to pay is the need for relatively long-range interactions.

Along these lines, **Joe Fitzsimons**, CEO of Horizon Quantum Computing, writes:

There are still research questions to be answered, so I would be hesitant to say that we know what the encoding [likely to drive the future evolution of fault-tolerant architectures] will be at present, but there is clearly a lot of progress being made on quantum LDPC codes.

Other proposals for fault tolerance that aim to reduce error-correction overhead are based on encoding quantum information into physical qubits that are inherently protected against certain types of error, such as qubits encoded in states of harmonic oscillators – so-called bosonic qubits. For such qubits, error-correction schemes can focus on the remaining error types.

Nicolas Menicucci explains why he sees bosonic codes as “the most promising scheme on paper”, while acknowledging that turning them into a working fault-tolerant architecture still requires some serious technological breakthrough:

Encoding a qubit in a single bosonic mode [...] is conceptually attractive because it lets you concentrate redundancy inside one well-controlled oscillator instead of spreading it over a large

“I believe that scaling to large numbers of logical qubits will only be possible with some form of hardware efficiency built into the error correcting code.

Without hardware efficiency, it is not just the qubit count that explodes, but the classical resources to control and measure these qubits.”

ALEXANDER BLAIS
University of Sherbrooke

array of bare qubits. [...] If we could reliably prepare and stabilise good bosonic code states [in superconducting architectures], they would make very natural “level-0” qubits on top of which more conventional stabiliser codes could sit.

[In some photonic proposals,] bosonic codes would live in optical modes rather than microwave cavities. There, the appeal is different – room-temperature operation and better prospects for large-scale integration.

[T]he core challenge is similar: we do not yet know how to make these encoded bosonic qubits with the fidelity and stability that fault tolerance really needs.

When it comes to physical implementations suitable for fault-tolerant quantum computing, **Lieven Vandersypen**, a professor at TU Delft, argues that spins in semiconductor quantum dots are promising because they are highly scalable due to integration. “Other schemes”, he comments, “will likely need many separate modules that are connected by optical or microwave links, which will likely become literally the weak links.”

KEY POINTS

- Research on quantum error-correction is a highly active field and could see significant breakthrough results.
- The choice of error-correction code may vary by architecture, based on, for example, the underlying physical errors or the kind of connectivity.
- Some variation of the surface code is still considered a leading choice for superconducting implementations.
- Quantum Low-Density Parity-Check (qLDPC) codes are actively being developed and adapted to hardware, because they reduce encoding overhead and facilitate scaling. They require long-range connectivity, though, which makes them not immediately suitable for all physical implementations.
- Bosonic codes also hold promise to reduce encoding overhead and facilitate scaling, but the sophisticated “0-level” encoding poses challenges.

4.4 Useful applications of early quantum computers

Developing quantum computers capable of compromising modern cybersecurity systems may still take considerable time (see Section 4.2). The development speed of these powerful machines largely depends on the funding allocated to quantum computing research. This funding may come from grants, venture capital, or revenue from early quantum technology applications and adoption. While there are various ways to attract investment and generate income, having commercially viable applications would significantly boost the likelihood of sustained financial support for advancing quantum computing to the level where it could influence cryptography. Therefore, we sought expert opinions on this by asking the following:

"I would think that quantum simulations in quantum chemistry and quantum materials that have actual commercial applications should be within reach in the next 5-10 years."

JENS EISERT
Freie Universität Berlin

Q: Please indicate your likelihood estimates for useful commercial applications of available processors – or of larger/less noisy processors but anyway not yet cryptographically-relevant – going beyond proof-of-concept and/or promotional activities, within the indicated timeframes.

The likelihood estimates provided by the respondents are summarized in Figure 23. Not all experts expressed an opinion, but among those who did, more than half indicated a likelihood of about 50% or more within 5 years.

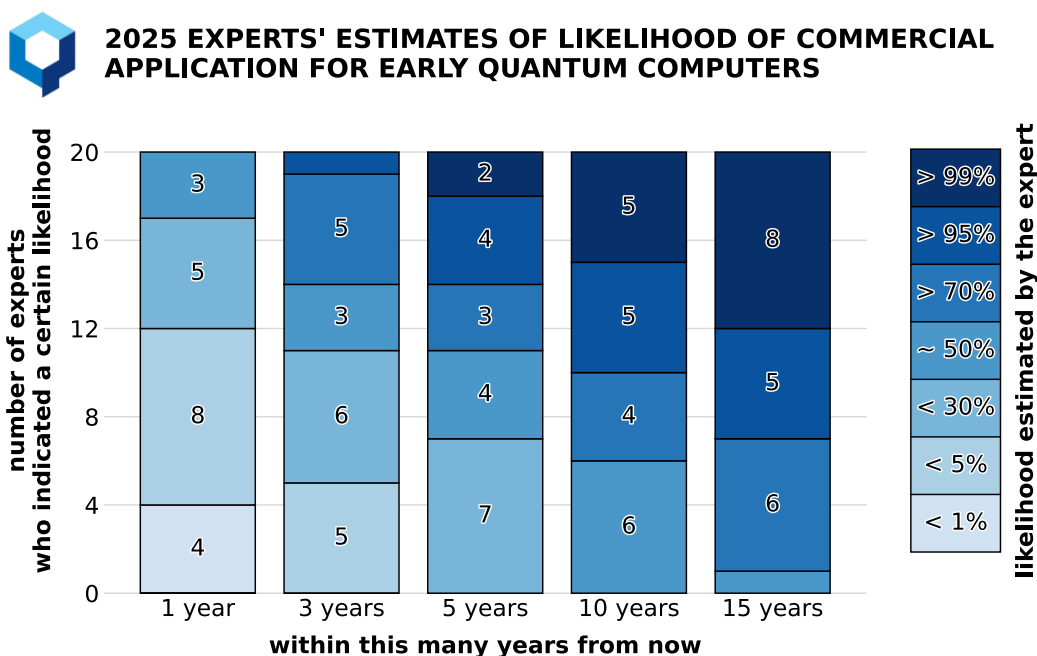


Figure 23 Likelihood estimates for commercial applications of "early" quantum computers / quantum processors not yet powerful enough to be directly relevant from a cryptographic perspective.

Several respondents spoke of quantum computers used for heuristic algorithms and simulations, and of scientific applications appearing before commercial applications. Among such respondents is **Daniel Gottesman**:

I think the short-term commercial applications will mostly be heuristic algorithms that we can't reliably test now (and may not have invented yet), leading to significant uncertainty. Small simulations are also possible in this time frame, but small simulations are mostly valuable for scientific research. To be commercially useful, bigger simulations are probably needed.

Some respondents stress that the notion of “commercial applications” is not well defined itself. One respondent writes:

I am taking the weakest definition: someone will be willing to pay some amount of money to perform a computation that classical computers will take longer to do. Under this very weak definition, I believe it is likely that, within the next year, quantum advantage will be reached for physical simulations, using error mitigation.

With respect to the general notion that early applications may contribute to the development of a cryptographically relevant quantum computer, another respondent raises two additional questions:

First, what will be the commercial applications of high-performance computing and AI in the next decade? (These technologies may draw funding and attention away from quantum computing.)

Second, what will be the commercial applications of other quantum technologies, such as quantum sensing? (Private sector investments here may indirectly support the development of quantum computers.)

KEY POINTS

- “Early” quantum computers may find uses long before it becomes possible to run standard quantum algorithms like Shor’s on them.
- The respondents mention a potential valuable role for such devices in running heuristic algorithms and conducting (research) simulations, especially in quantum materials and quantum chemistry.
- When it comes to heuristic algorithms and simulations, it is difficult to definitively assert there is a quantum advantage, as, for example, there might be unknown classical approaches that could perform similarly or better.
- It might be necessary to “play” with early quantum computers as they become available, to understand what tasks or subtasks they may be helpful for.

4.5 `Under the radar` research

There may be ongoing research efforts that remain undisclosed, such as those conducted by government-supported laboratories, companies operating in stealth mode, or potentially malicious private entities. We asked the respondents to share their thoughts on how substantial the impact of such covert research activities could be on the timeline for creating a cryptographically relevant computer. We remark that, when providing their likelihood estimates (see Section 4.2), the experts could have already included such an effect.

The experts could estimate how many years the timeline might shift, choosing from a set of options. Here, we report the statistics of the 22 responses we received on the topic (Figure 24). While about a third of the experts suggested that the effect is nonexistent or very limited (up to one year), a majority indicated that covert research could shift the timeline by two or more years. This implies that, although this effect should be considered as ‘diluted’ in the context of potentially long timelines—after all, the likelihood estimates extend to 30 years (see Section 4.2)—extra caution is warranted when evaluating quantum risk: since covert successes would likely remain invisible for some time, it is safer to assume that the true threat could be closer than what can be inferred from open publications alone.

“At present, I do not expect covert activities to be significantly ahead of publicly known activities. As the moment comes closer, it is possible for companies to go partially in stealth mode, working alongside state-backed labs, for a focused effort in code-breaking.”

LIEVEN VANDERSYPEN
TU Delft

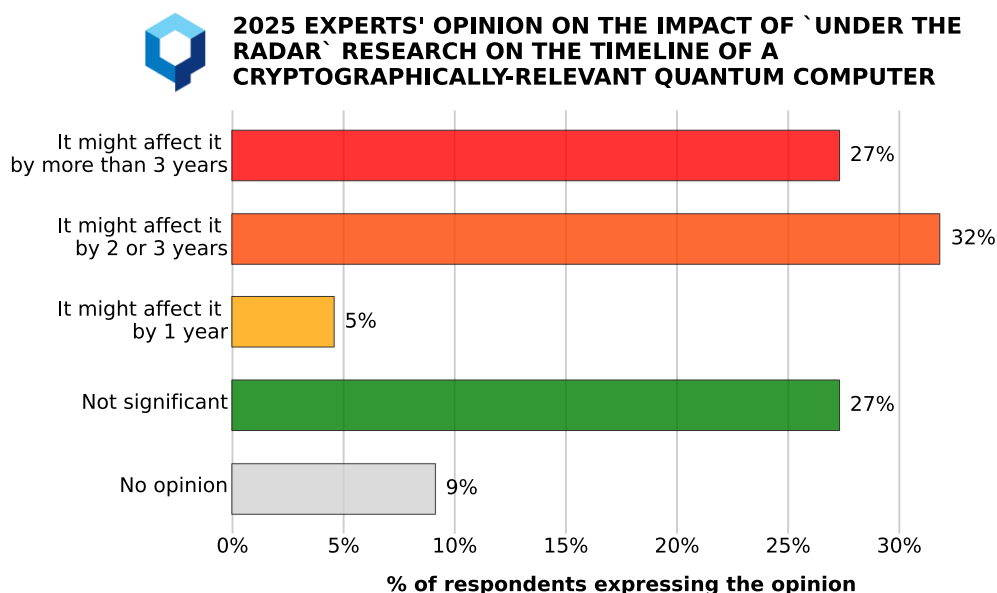


Figure 24 Twenty-two respondents shared their intuition about the impact covert research may have on the quantum threat timeline.

Some comments of the respondents point at any covert research as to something that widens the uncertainty around the timeline. **Nicolas Menicucci** wrote:

Stealthy or opaque efforts mostly widen the uncertainty window rather than shifting my central estimate.

Among those expecting little or no effect to the quantum threat timeline, **Yvonne Gao** explained:

Doing [quantum computing research] right takes so much collective effort from so many different fields that I highly doubt any entity can do so fully under the radar. No single entity would have the depth and breadth of knowledge that can sustain such developments purely on their own.

On the other hand, one respondent cautioned:

Looking at the historical development of cryptanalysis and the development of earlier technologies such as microwave electronics and radar (which are not entirely unrelated to quantum), it is certainly possible for significant discoveries to occur and then be kept secret.

Finally, some respondents emphasized the importance and effectiveness of open research; **Jens Eisert** mentioned his belief that

Open research is so much faster than undisclosed research efforts will have a hard time catching up.

KEY POINTS

- Covert research, such as that conducted by government-supported laboratories, by companies operating in stealth mode, or by malicious private entities, may influence the quantum threat timeline in significant ways.
- The experts overall agree that this is a concrete possibility, even though quantum computing research is challenging enough that several experts do not think covert research can progress much faster than open research. Similarly, some other experts believe it would be exceedingly difficult to maintain any significant research effort in secret.
- Extra caution is warranted when evaluating quantum risk: since covert successes would remain invisible for some time, it is safer to assume that the true threat could be closer than what can be inferred from open publications alone.

4.6 Societal and funding factors

This section presents the results of questions evaluating how societal and funding factors might influence the timeline for developing a cryptographically relevant quantum computer.

4.6.1 Level of funding of quantum computing research

Long-term investments are essential to advance the development of a fully fault-tolerant quantum computer. As global leaders in this field—engaged in both national and international projects, collaborating with industry, and leading start-ups—our respondents possess a unique perspective for assessing the trajectory of funding.

“The close relationship between academia and the quantum industry should be maintained.”

JENS EISERT
Freie Universität Berlin

As we have done since 2020, we have asked them to forecast what is likely to happen regarding funding in the next two years. The results of the 2025 survey are presented in Figure 25 alongside earlier results. We note that the experts were asked to comment on overall funding, including venture-capital funding, funding within large corporations, and public support (Kung and Fancy 2021).

The results of this year don’t differ too much from those of last year, suggesting renewed investments, compared to a slowdown signal in 2023. A majority of respondents think there will be an increase, and some expect a substantial one. Based on the comments of both last year and this year, this is related to recent progress in the field. This year, one respondent commented:

We are now at the beginning of a steep rise in interest in quantum, since the confidence in the fact that large-scale quantum computing will indeed become a reality has greatly increased due to the past year's error-correction experiments. These experiments [...] demonstrated that quantum computing beyond the fault tolerance break-even point is possible. This means that we "only" need to increase the number of high-quality qubits.

Nonetheless, this year also sees the largest “will decrease” percentage since we started asking this question. **Bill Coish** provided a rationale:

My expectation is that most of the recent funding/capital that has been deployed is predicated on a rapid expansion of quantum computing, but I believe all platforms face serious engineering/practical bottlenecks that will most likely delay rapid scaling, leading to a small decrease in funding in the short term (next two years).

Daniel Gottesman comments on the uncertainties that may affect both public and private investments:

U.S. funding under the current administration is highly uncertain and vulnerable to rapid unpredictable changes. I think government funding worldwide may increase due to a quantum computing arms race, but industry funding may [...] decrease as some companies that have dipped their toe into investigating quantum computing pull back and wait for more progress.

Nicolas Menicucci warns about the potential negative role of hype (see also Section 4.6.3):

My concern is that a lot of current quantum investment, especially on the private side, has a strong hype component. In the short term, this can accelerate progress by attracting capital and people. But if

expectations are set too high, the eventual disappointment when promised milestones fail to materialise can trigger a “quantum winter” in which funding and political support fall sharply. That would slow progress overall, even if the underlying science is healthy.

The results of our surveys over the past years have been roughly in line with the trends in investment figures analyzed by McKinsey & Company. In their most recent Quantum Technology Monitor, released in June 2025 (McKinsey & Company 2025), they report a ~50% increase in investments in quantum-technology start-ups in 2024, with the vast majority directed towards quantum computing, matching the positive expectations expressed by our respondents in 2024. McKinsey & Company points out that the increase was particularly significant for public investments. This likely reflects recognition of the



OVER THE NEXT TWO YEARS, THE LEVEL OF GLOBAL INVESTMENT (BOTH BY GOVERNMENT AND BY INDUSTRY) TOWARDS QUANTUM COMPUTING WILL ...

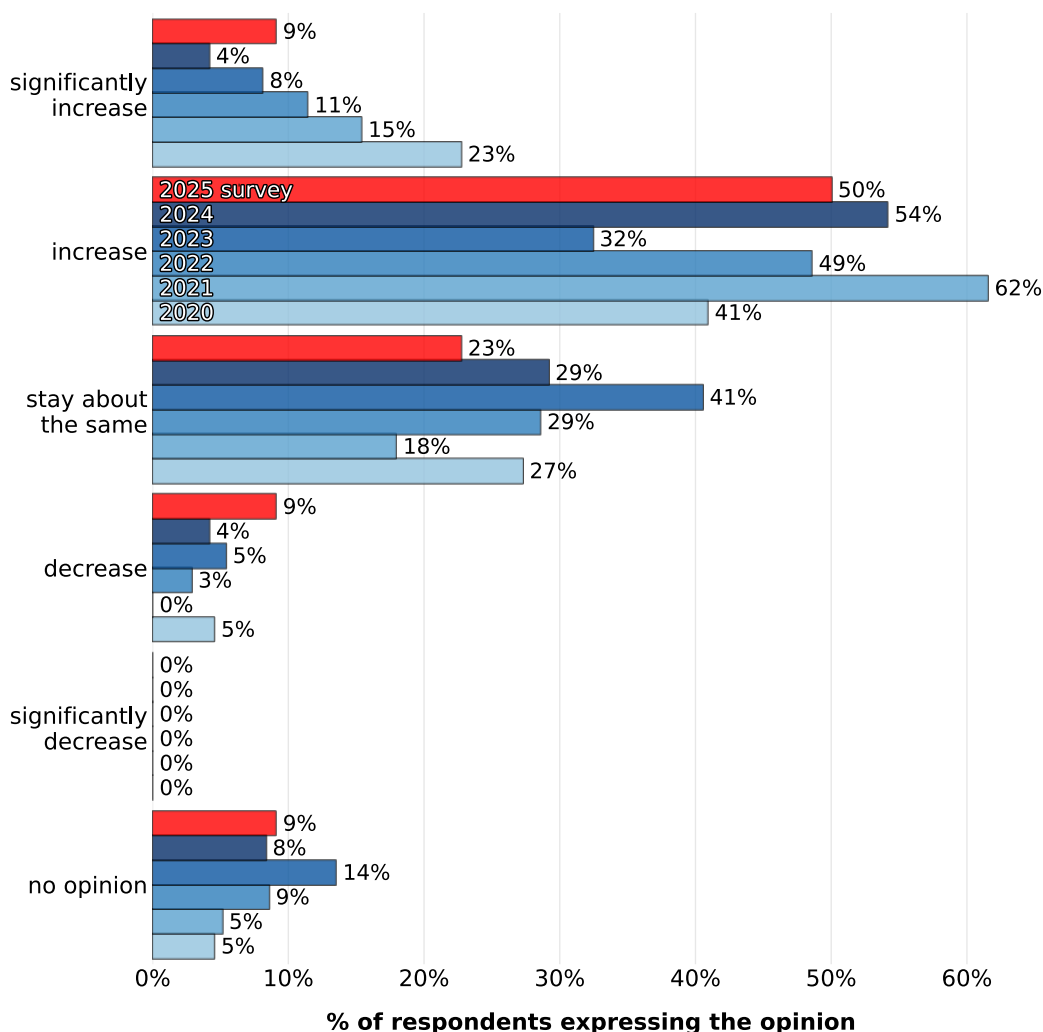


Figure 25 Expected shift in investment levels toward quantum computing over the next two years, comparing estimates for the six surveys from 2020 to 2025. The number of responses varied from survey to survey, and we report percentages.

strategic importance of quantum technologies, particularly quantum computing (see Section 4.6.2), and awareness of the field's recent significant progress.

4.6.2 Global race to build a fault-tolerant quantum computer

The pursuit of a quantum computer with cryptographic significance is akin to a race on multiple fronts. In Section 4.1, we considered the “rivalry” among various architectures. In this section, our focus is on the contest involving both national and supranational entities, such as the European Union.

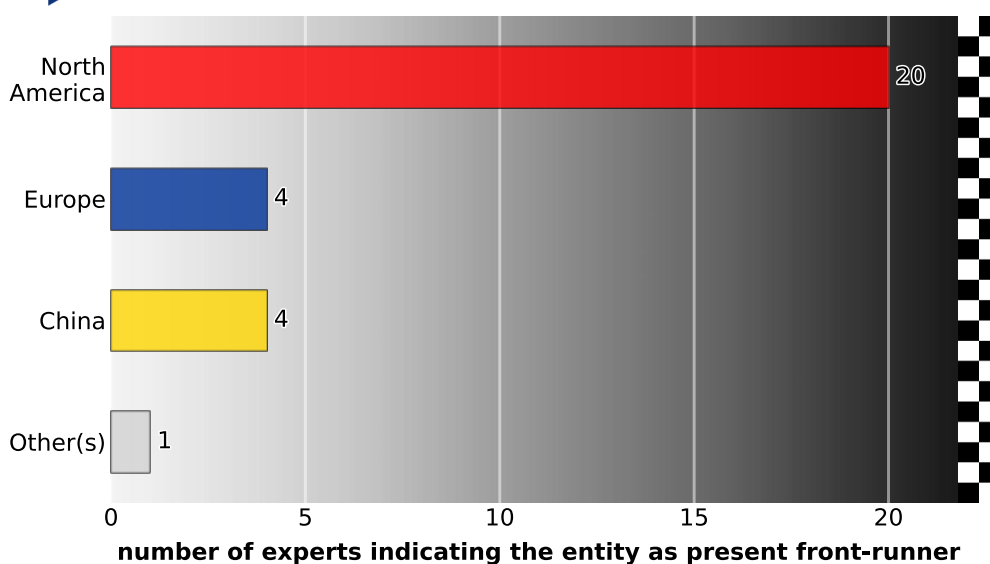
Many nations actively recognize the successful creation of a quantum computer as a strategic objective (Kung and Fancy 2021). This is because such a development would revolutionize not just cryptography and much of our digital framework — the primary focus of this report — but also various societal and economic sectors. For instance, consider the potential to efficiently emulate quantum systems when developing innovative materials and drugs.

“There are examples of excellent initiatives in Europe and China, but the center of activity with the most promising companies is still the United States.”

BILL COISH
McGill University

This underlying rivalry is a significant driver of investment in the quantum computing sector. As a result, tracking the progress and potential directions of this “race” offers valuable insights into the timeline of the quantum threat. Additionally, for those responsible for mitigating the quantum threat, it is essential to identify where it may originate. This involves determining which entities are most likely to develop a quantum computer capable of breaking current encryption.

2025 EXPERTS' OPINIONS ON PRESENT FRONT-RUNNERS IN THE "GLOBAL RACE" TO BUILD A QUANTUM COMPUTER



Note: Each expert could select multiple choices.

Figure 26 Number of respondents that indicated a region/entity as present front-runner in the global race to build a fault-tolerant quantum computer. Twenty-one experts provided input; multiple answers were allowed.

We solicited expert opinions to identify which regions among China, Europe, and North America are currently leading, allowing for multiple responses and the inclusion of other regions⁸. Twenty-one experts provided an opinion. The results are shown in Figure 26.

According to those who answered with specific choices, North America appears to be the present leading world region by far, followed by Europe and China, with the latter two ranked similarly. In the last two recent surveys, North America has substantially strengthened its present leadership. While this might in part be an effect of the different composition of the respondent pool, it is also likely driven by the recent strong results in quantum error correction and logical encoding by U.S.-based companies and research teams.

Given our interest in future trends, we also asked the experts to indicate the likelihood of each region previously considered to be a frontrunner five years from now, and whether new frontrunners may emerge. The results are presented in Figure 27. Most respondents consider it likely that North America will maintain its frontrunner position, but will be challenged by China. Europe appears to lag in expectations, and many respondents consider it unlikely that it will have the status of a frontrunner in five years. This is particularly striking given that, at face value, Europe is currently considered to be on par with China (Figure 26).

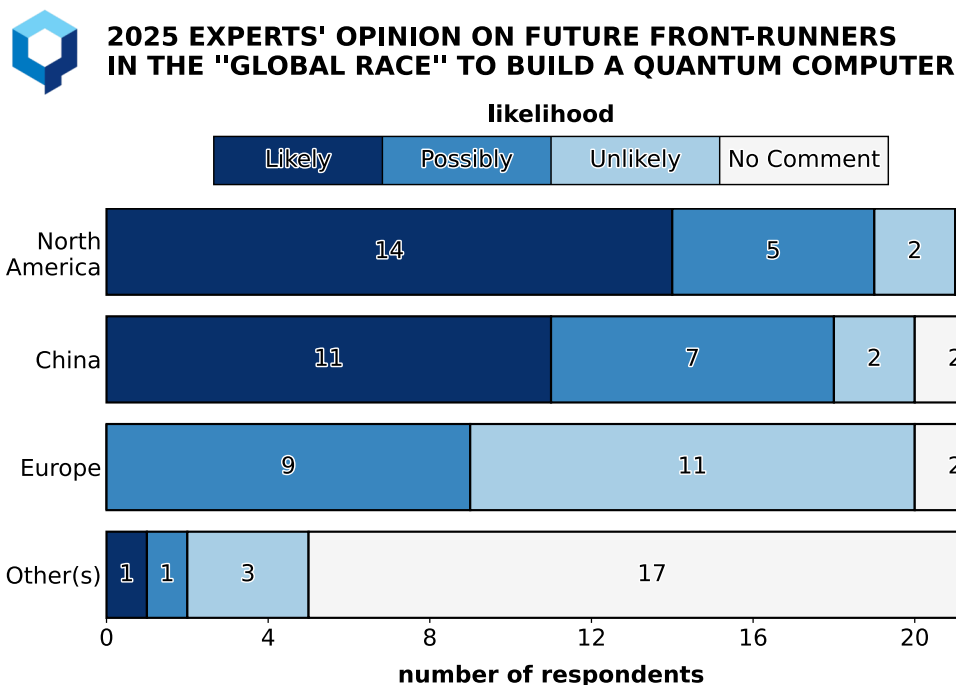


Figure 27 Number of respondents that indicated the likelihood of a given region/entity to be a front-runner in the global race to build a fault-tolerant quantum computer, five years from now.

⁸ The reader may consider taking into account the geographical composition of our pool of respondents (see Section 3).

Daniel Gottesman's comment explains why this might be the case:

Europe doesn't seem to have the same investment in quantum computers from large companies that we see in North America, and while government funding is strong, it is not at the same level of a major commitment that we see in China. It is difficult for small startups or academic groups to compete with a large company putting major resources into building a quantum computer. A focused government lab might.

Stephanie Simmons's comment is similar in spirit to Gottesman's – like also the comments from several other experts – but further stresses the specific role of the U.S. Quantum Benchmarking Initiative (QBI) program:

The US has ensured it is a front runner via its QBI program, regardless of the QBI participants' HQ location. All of the publicly remaining DARPA QBI efforts are from the five eyes countries – Europe has immense talent but is no longer in the global quantum shortlist. It is likely that China has matching – or larger – efforts underway.

When it comes to “Other(s)” countries, Australia and Japan are mentioned as having potential, particularly due to their strengths in quantum research. Canada – included in “North America” – is seen by some as an important player, but in a delicate position when it comes to securing its future status in quantum computing research.

4.6.3 Ethical concerns

We would like to understand the extent to which experts in the field think ethical concerns could influence the timeline for realizing a cryptographically relevant quantum computer, potentially through regulatory capture meant to mitigate them.

In Figure 28, we report how the respondents answered with respect to some specific possible ethical concerns, including quantum hype, global tensions spurred by a quantum “arms race”, endangering privacy, and (in)equitable access to technology.

Some respondents note that the quantum “arms race” is already leading to potential slowdowns, for example via export controls. Nonetheless, it is clear that the major concern is the role of hype. **Nicolas Menicucci** comments:

Over-promising on timelines and applications is not just harmless optimism; it sets up a backlash cycle. [...] In the short term, hype can attract capital; in the longer term, it risks slowing progress by provoking a funding retreat just when the underlying science is ready to pay off.

On the other hand, a respondent explains why hype is not a concern for them:

I am not concerned by hype as a factor slowing things down. It's annoying and often cringeworthy from a scientist's point of view, but it is also normal part of the business development cycle. Nothing new here.



2025 EXPERTS' OPINION ON ETHICAL CONCERNS POTENTIALLY AFFECTING THE REALIZATION OF A CRYPTOGRAPHICALLY RELEVANT QUANTUM COMPUTER

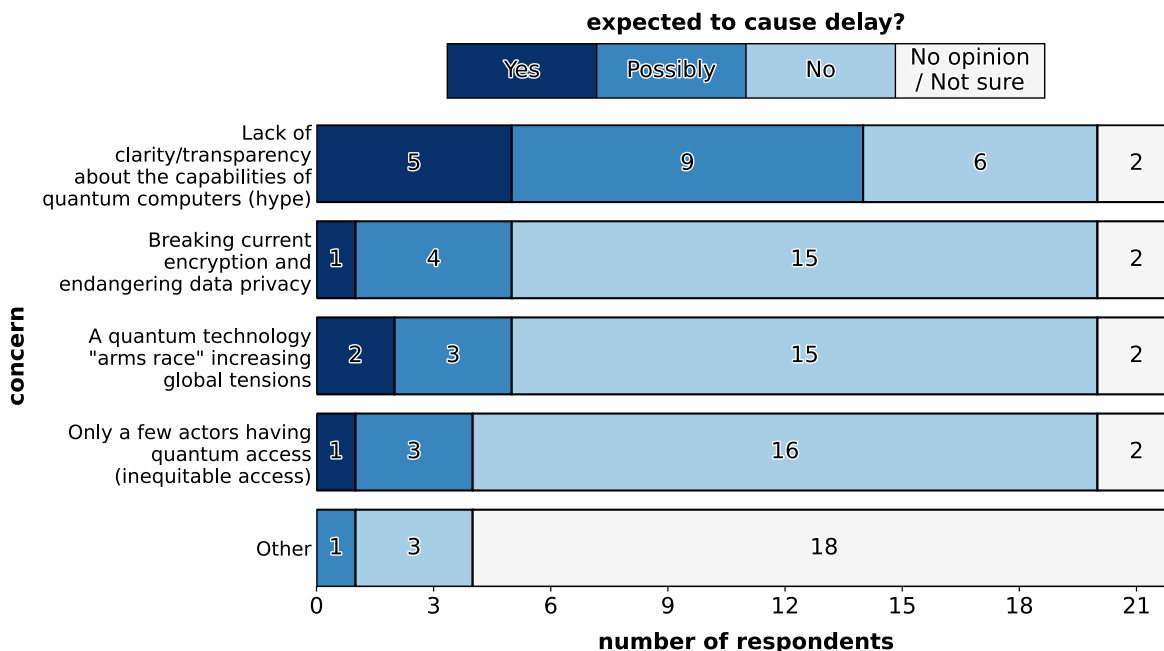


Figure 28 The experts shared their view about the possibility that ethical concerns may slow down quantum computing research.

KEY POINTS

- The journey towards realizing a quantum computer is often termed the 'quantum race'. Competition exists at both the geopolitical and the private company levels.
- Investments in quantum computing research influence the pace of development. Generally, experts expect investment in this field to keep growing, also fueled by recent results.
- At the geopolitical level, North America is seen as currently leading the 'race', with China considered a contender, particularly in a 5-year timeframe. Europe is perceived as falling behind. Some other countries, like Australia and Japan, are also considered to have significant potential.
- There are ethical concerns regarding quantum technologies and, in particular, quantum computing. Generally, our respondents think that they will not quite impact the pace of development. The main exception is a concern for the role that hype may have in leading to a 'quantum winter' of greatly reduced investments and research effort.

4.7 Current progress

In this section, we present opinions about the status of progress in quantum computing research and development.

4.7.1 Recent developments

The latter half of 2024 and early 2025 have witnessed a surge in the achievement of experimental milestones, particularly in the performance of error-corrected logical qubits compared to their physical counterparts. Respondents emphasized that the field is moving beyond simple break-even demonstrations toward robust error suppression⁹.

A major highlight identified by multiple experts is the advancement in superconducting architectures, specifically Google's results with the "Willow" processor (Acharya et al. 2025). Experts noted that this processor has demonstrated quantum advantage and, crucially, performance beyond the break-even point for error correction. One respondent detailed that the processor realized distance-5 and distance-7 surface-code memories, with the distance-7 logical qubit surviving more than twice as long as the best physical qubit.

Beyond superconducting circuits, significant progress was noted in other implementations.

Experts described progress in neutral atoms as "truly spectacular," particularly the activities surrounding Rydberg atom arrays (see, for example, (Bluvstein et al. 2025; Manetsch et al. 2025)). However, it was noted that the "very small clock rate" remains a challenge for running complex algorithms on this platform. The progress in error correction and logical computation in ion-trap platforms was also considered a recent experimental highlight (see, for example, (Daguerre et al. 2025)).

The community reacted positively to the experimental concatenation of bosonic cat qubits in a repetition code (Putterman et al. 2025). This was highlighted as an excellent example of exploiting noise bias to create a pathway for low-overhead quantum error correction.

On the theoretical side, experts pointed to (Gidney 2025) as a candidate for the most significant achievement, noting it reduces the number of qubits required to factor a 2048-bit number to below 1 million. This makes the timeline for cryptographically relevant quantum computers more plausible than estimates from several years ago. Another work highlighted as significant along these lines is (Chevignard et al. 2025).

4.7.2 Next near-term step

When asked what essential steps should be achieved by roughly Summer 2026, the consensus shifted from establishing logical memory to demonstrating logical computation and scalability.

A key milestone would be a programmable logical processor with 5–10 logical qubits capable of executing a universal gate set—including fault-tolerant non-Clifford gates—that outperforms unencoded implementations.

⁹ Given the timing of the announcements of such results and of our survey, not all the experts taking part in our survey were aware of the same set of results.

There is a strong call for more efficient codes with a smaller ratio of physical to logical qubits. Specifically, respondents want to see progress in the realization of quantum Low-Density Parity-Check (qLDPC) codes and bosonic GKP codes.

A recurring theme for the next year is the demonstration of "horizontal scale". Experts are looking for coherent processing across multiple chips or cryostats, effectively distributing a quantum computer across distant nodes.

KEY POINTS

- The past two years have seen noteworthy progress, with impressive demonstrations of quantum error correction and logical encoding, in multiple types of architectures.
- Progress is expected to continue in such a direction in the near future, moving from logical memories to logical computation.
- Two complementary ways to improve scalability of quantum devices are the development of more efficient error-correcting codes and coherently distributing computation across multiple devices.

4.8 Sources of unexpected speed-up

Advancements in quantum computing research could significantly and unexpectedly speed up the development of a CRQC. The field includes various subfields, making it important to identify which areas have the greatest potential for breakthroughs. Understanding this can also help track overall progress.

To gather insight, we asked respondents to share their opinions on which aspects of quantum computing research are most likely to yield significant, unforeseen progress. The results are shown in Figure 29. The respondents identified improvements in quantum error correction schemes and in hardware as the most promising areas.

Daniel Gottesman provides a short overview of the potential of each subfield as he perceives it:

"Much can be done to speed up [the creation of a] cryptographically relevant quantum computer. The current schemes and algorithms are highly unlikely to be the best possible..."

"[...] Developments in AI are highly likely to introduce new components into the equation."

RESPONDENT

Modular architecture and compilation [...] are important, but I don't see big surprises coming from there. Modular architecture is more of a risk than a potential source of great progress; many systems depend on it and if it turns out to be extremely difficult, that could substantially slow progress compared to the level of improvements we are seeing now. I think there is still significant room for improvement in fault-tolerant protocols by using new ideas. For quantum algorithms, I expect to see a boom in new heuristic algorithms once we have reliable quantum computers large

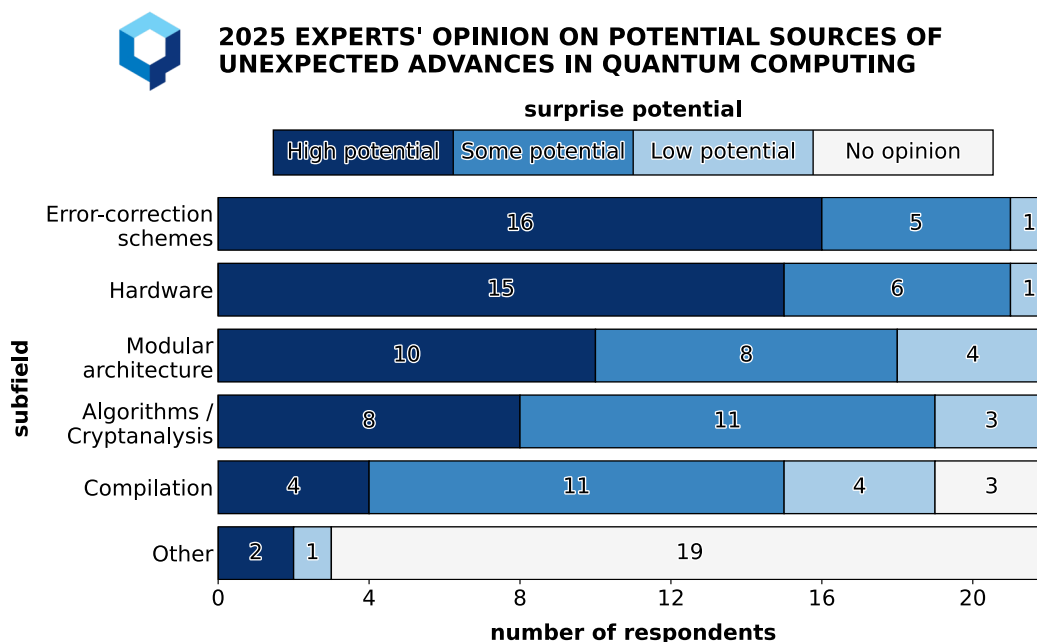


Figure 29 Number of respondents that indicated a certain "surprise potential" for several subfields of quantum computing research, which could also be seen as "layers" of a quantum stack.

enough to test a variety of ideas, but it is difficult to predict what these heuristic algorithms will do or whether they will be useful enough to spur significant increased investment.

It is worth stressing that, while Gottesman points to modularity as being a risk factor, he nonetheless mentions its importance. The role of modularity is further stressed by other respondents, like **Bill Coish**:

I think the major bottlenecks right now are in engineering very high-quality quantum hardware. A key requirement for most implementations will also be to create a modular architecture with the associated hardware requirements for sharing entanglement across interconnects. Improved compilation schemes have led to some of the biggest (theoretical) improvements in resource requirements.

KEY POINTS

- There can be significant and unexpected accelerations in quantum computing research.
- The experts identified quantum error correction as well as hardware development as the subfields of research more likely to produce such accelerations.
- Many respondents point to modularity as necessary to scale to a full-fledged fault-tolerant quantum computer that is cryptographically relevant, despite coming with its own challenges.

Summary and outlook

A quantum computer capable of effectively running quantum cryptanalysis algorithms poses a threat to widely adopted cryptosystems that rely on computational problems believed to be intractable for current computers but relatively easy for a cryptographically relevant quantum computer (CRQC).

Building a CRQC will require years of progress in both science and engineering, which can only be achieved through dedicated commitment and sufficient resources. The main challenge to address is the inherent ‘fragility’ of the quantum features that give quantum computing its advantage over classical computing. The end goal of quantum computing research is computations that use logical qubits, a dependable way to encode and handle quantum information, even when the underlying physical qubits are prone to errors.

The expert opinions collected and summarized in this report offer unique insights into the timeline of the quantum threat. Twenty-six respondents estimated the likelihood of a quantum computer capable of breaking a scheme like RSA-2048 within 24 hours – our working notion of a CRQC – and shared their opinions on various aspects of quantum computing development and on diverse factors influencing it.

The past year has seen several experimental demonstrations of quantum error correction, logical encoding, and logical computation that our respondents generally find impressive. Alongside advances in error-mitigation and correction techniques, increases in the number of physical qubits across platforms, and the theoretical development of efficient error-correction schemes, these achievements support an optimistic outlook for the future of quantum computing.

Recent advancements have been so significant that this year’s opinions are the most optimistic for the “within 10 years” timeframe that we have ever recorded in our series of surveys: half of the respondents

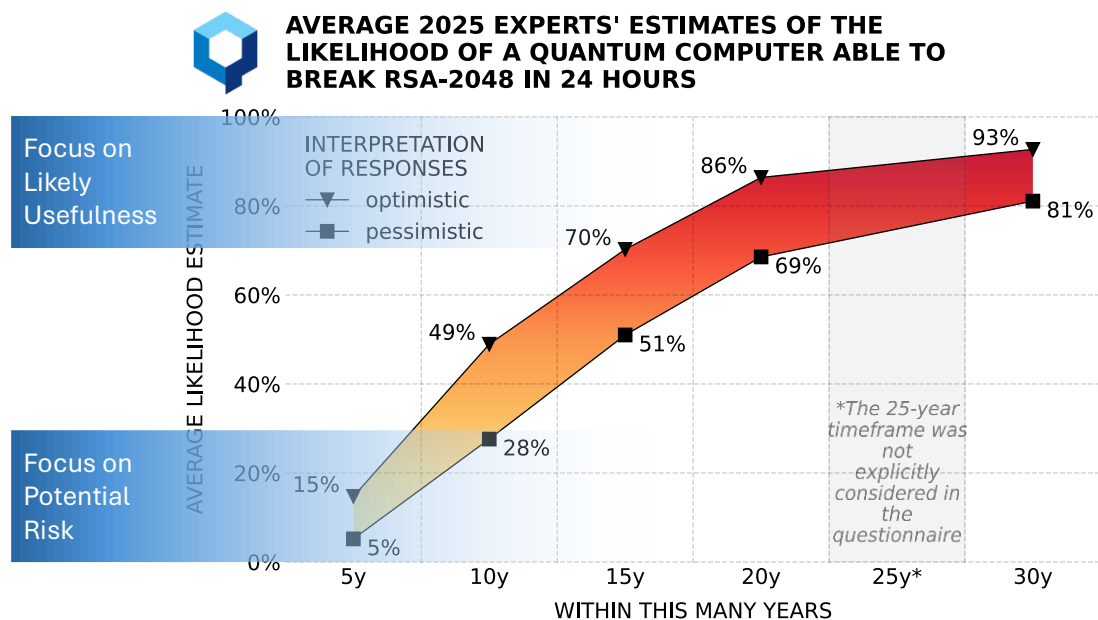


Figure 30 Average expert likelihood estimates of a cryptographically relevant quantum computer (same as Figure 3) with added graphics highlighting the different focus one can adopt when looking at the chart: from a utility perspective, or from a risk perspective.

(13/26) felt the likelihood of such a computer within 10 years was “about 50%” or more likely. By coarse-graining all the experts’ responses, one arrives at an average likelihood of between 28% and 49% within 10 years, that is, by roughly 2035 (see Figure 30 for reference). When one considers the years our surveys were conducted in, this year’s likelihood estimates for ~2035 aren’t the highest recorded, but we think that this should not distract from the impressive recent pace of progress, for at least two reasons: firstly, the pandemic and the financial scenario immediately following the pandemic may have acted as a watershed period that set a new baseline for the expectations related to the timing of quantum computing development; secondly, the likelihood estimates of the current survey are the highest among the surveys for the most recent four years.

The graphics of Figure 30 also stress that the part of the likelihood curve one may choose to focus on depends, for example, on whether one is interested in near-certain positive applications of quantum computing, or in the cryptographic risk associated with quantum computing. That is, while one could dismiss quantum computers as unlikely to run proper, useful quantum algorithms for many years, cyber-risk managers are naturally more concerned about the chance that the quantum threat materializes early, and potentially earlier than many could expect, rather than never. There is no doubt that, while the risk aversion/appetite of companies and institutions can vary significantly, the estimated likelihoods represent a very serious concern for critical systems.

There are several reasons why actions to manage the quantum threat should start earlier rather than later. First, one should consider that malicious agents may adopt a “Harvest Now, Decrypt Later” (HNDL) approach, storing valuable encrypted data waiting for a CRQC to become a reality. Second, some of the progress being made is not fully visible to the global academic community, and this trend is likely to grow in the coming years. Significant advancements will continue to emerge out of sight and remain partially concealed. Third, slow-rolling preparations against the quantum threat set the conditions for at least two kinds of dangers (García 2025). For one, a hasty transition to quantum-safe tools may suddenly become a forced choice, with all the risks associated with it, from a breakdown of services to involuntarily creating vulnerabilities even against more traditional attacks. In addition, if the transition to quantum-safe solutions is so slow/delayed that it is not complete by the time a CRQC becomes available, the integrity and authenticity of information, the availability of connected systems, and the control of connected devices will be at risk in real time.

Cyber solutions that can start to be implemented are available today (World Economic Forum 2023; Canadian Forum for Digital Infrastructure Resilience 2024). This is facilitated by the US

“It is important to stress — not least given the roadmaps presented by industry — the importance of migrating to post-quantum secure cryptography. In particular, this is important in applications where long-term confidentiality is sought. This is because adversaries can store ciphertexts that are intercepted now for decryption sometime in the future when large-scale fault-tolerant quantum computers become available.”

RESPONDENT

Cyber-risk managers should monitor advancements in the experimental development of quantum error correction to gauge how rapidly cryptographically relevant quantum computers are emerging.

National Institute of Standards and Technology (NIST) having recently issued the first standards for post-quantum cryptographic algorithms (NIST, 2024). The Global Risk Institute and evolutionQ Inc. have made available an updated Quantum Risk Assessment (QRA) methodology (Mosca and Mulholland 2025) on which such a process may be based.

Cybersecurity experts and managers should also monitor developments in quantum computing research, demonstrations, and commercialization to assess the pace of quantum computer advancement. They should further consider the possibility of advancements in cryptanalysis algorithms, which would enable cryptanalysis with fewer quantum resources – say, fewer quantum qubits, or fewer computational steps – than the current state of the art. Most importantly, unexpected advances in cryptanalysis may also threaten the security of cryptographic algorithms that are immune to the known quantum attacks, which is one further reason for persistent and increasing calls for cryptographic resilience enabled by cryptographic agility, reliable alternative cryptographic schemes, and cryptographic defense-in-depth.

References

- Acharya, Rajeev, Dmitry A. Abanin, Laleh Aghababaie-Beni, et al. 2025. "Quantum Error Correction below the Surface Code Threshold." *Nature* 638 (8052): 920–26. <https://doi.org/10.1038/s41586-024-08449-y>.
- Arute, Frank, Kunal Arya, Ryan Babbush, et al. 2019. "Quantum Supremacy Using a Programmable Superconducting Processor." *Nature* 574 (7779): 7779. <https://doi.org/10.1038/s41586-019-1666-5>.
- Barnes, Itan, Colin Soutar, and Alvaro Veliz Osorio. 2025. *Why Quantum Computers Aren't Cracking RSA Yet*. Deloitte.
- Bluvstein, Dolev, Alexandra A. Geim, Sophie H. Li, et al. 2025. "Architectural Mechanisms of a Universal Fault-Tolerant Quantum Computer." arXiv:2506.20661. Preprint, arXiv, June 25. <https://doi.org/10.48550/arXiv.2506.20661>.
- Breuckmann, Nikolas P., and Jens Niklas Eberhardt. 2021. "Quantum Low-Density Parity-Check Codes." *PRX Quantum* 2 (4): 040101. <https://doi.org/10.1103/PRXQuantum.2.040101>.
- Canadian Centre for Cyber Security. 2025. "Roadmap for the Migration to Post-Quantum Cryptography for the Government of Canada (ITSM.40.001)." Canadian Centre for Cyber Security, June 24. <https://www.cyber.gc.ca/en/guidance/roadmap-migration-post-quantum-cryptography-government-canada-itsm40001>.
- Canadian Forum for Digital Infrastructure Resilience. 2024. *Canadian National Quantum-Readiness: Best Practices and Guidelines*. Canadian Forum for Digital Infrastructure Resilience (CFDIR). <https://ised-isde.canada.ca/site/spectrum-management-telecommunications/sites/default/files/documents/Quantum-Readiness%20Best%20Practices%20-%20v04%20-%2010%20July%202024.pdf>.
- Chevignard, Clémence, Pierre-Alain Fouque, and André Schrottenloher. 2025. "Reducing the Number of Qubits in Quantum Factoring." In *Advances in Cryptology – CRYPTO 2025*, edited by Yael Tauman Kalai and Seny F. Kamara. Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-01878-6_13.
- Cochrane, P. T., G. J. Milburn, and W. J. Munro. 1999. "Macroscopically Distinct Quantum-Superposition States as a Bosonic Code for Amplitude Damping." *Physical Review A* 59 (4): 2631–34. <https://doi.org/10.1103/PhysRevA.59.2631>.
- Daguerre, Lucas, Robin Blume-Kohout, Natalie C. Brown, David Hayes, and Isaac H. Kim. 2025. "Experimental Demonstration of High-Fidelity Logical Magic States from Code Switching." *Physical Review X* 15 (4): 041008. <https://doi.org/10.1103/dck4-x9c2>.
- DARPA. 2025. "QBI: Quantum Benchmarking Initiative | DARPA." DARPA. <https://www.darpa.mil/research/programs/quantum-benchmarking-initiative>.
- DiVincenzo, David P. 2000. "The Physical Implementation of Quantum Computation." *Fortschritte Der Physik* 48 (9–11): 771–83. [https://doi.org/10.1002/1521-3978\(200009\)48:9/11%253C771::AID-PROP771%253E3.0.CO;2-E](https://doi.org/10.1002/1521-3978(200009)48:9/11%253C771::AID-PROP771%253E3.0.CO;2-E).
- Eisert, Jens, and John Preskill. 2025. "Mind the Gaps: The Fraught Road to Quantum Advantage." arXiv:2510.19928. Preprint, arXiv, October 22. <https://doi.org/10.48550/arXiv.2510.19928>.
- Feynman, Richard P. 1982. "Simulating Physics with Computers." *International Journal of Theoretical Physics* 21 (6): 467–88. <https://doi.org/10.1007/BF02650179>.
- Fowler, Austin G., Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. 2012. "Surface Codes: Towards Practical Large-Scale Quantum Computation." *Physical Review A* 86 (3): 032324. <https://doi.org/10.1103/PhysRevA.86.032324>.

- García, Jaime Gómez. 2025. "Crypto-Procrastination: Preparing for a Quantum Secure Economy, Today." *Infosecurity Magazine*, July 1. <https://www.infosecurity-magazine.com/opinions/crypto-procastination-quantum/>.
- Gheorghiu, Vlad, and Michele Mosca. 2025. "Quantum Resource Estimation for Large Scale Quantum Algorithms." *Future Generation Computer Systems* 162 (January): 107480. <https://doi.org/10.1016/j.future.2024.107480>.
- Gidney, Craig. 2025. "How to Factor 2048 Bit RSA Integers with Less than a Million Noisy Qubits." arXiv:2505.15917. Preprint, arXiv, May 21. <https://doi.org/10.48550/arXiv.2505.15917>.
- Gidney, Craig, and Martin Ekerå. 2021. "How to Factor 2048 Bit RSA Integers in 8 Hours Using 20 Million Noisy Qubits." *Quantum* 5 (April): 433. <https://doi.org/10.22331/q-2021-04-15-433>.
- Google Quantum AI. "Google Quantum AI Roadmap." <https://quantumai.google/roadmap>.
- Gottesman, Daniel, Alexei Kitaev, and John Preskill. 2001. "Encoding a Qubit in an Oscillator." *Physical Review A* 64 (1): 012310. <https://doi.org/10.1103/PhysRevA.64.012310>.
- IBM Quantum Roadmap. "IBM Quantum Roadmap." <https://www.ibm.com/quantum/hardware>.
- IonQ Roadmap. "IonQ Roadmap." <https://www.ionq.com/roadmap>.
- Kitaev, A. Yu. 2003. "Fault-Tolerant Quantum Computation by Anyons." *Annals of Physics* 303 (1): 2–30. [https://doi.org/10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0).
- Kung, Johnny, and Muriam Fancy. 2021. "A Quantum Revolution: Report on Global Policies for Quantum Technology." CIFAR, April 7. <https://cifar.ca/wp-content/uploads/2021/05/QuantumReport-EN-May2021.pdf>.
- Manetsch, Hannah J., Gyohei Nomura, Elie Bataille, Xudong Lv, Kon H. Leung, and Manuel Endres. 2025. "A Tweezer Array with 6,100 Highly Coherent Atomic Qubits." *Nature* 647 (8088): 60–67. <https://doi.org/10.1038/s41586-025-09641-4>.
- McKinsey & Company. 2025. *Quantum Technology Monitor 2025 | McKinsey*. <https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/the-year-of-quantum-from-concept-to-reality-in-2025>.
- Microsoft Quantum Roadmap. "Microsoft Quantum Roadmap." <https://quantum.microsoft.com/en-us/vision/quantum-roadmap>.
- Mosca, Michele. 2013. *E-Proceedings of 1st ETSI Quantum-Safe Cryptography Workshop*.
- Mosca, Michele, and John Mulholland. 2025. "An Updated Methodology for Quantum Risk Assessment." Global Risk Institute. <https://globalriskinstitute.org/publication/an-updated-methodology-for-quantum-risk-assessment/>.
- Nielsen, Michael A., and Isaac L. Chuang. 2000. *Quantum Computation and Quantum Information*. Cambridge University Press.
- NIS Cooperation Group. 2025. "A Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography | Shaping Europe's Digital Future." <https://digital-strategy.ec.europa.eu/en/library/coordinated-implementation-roadmap-transition-post-quantum-cryptography>.
- NIST. 2024. "Announcing Issuance of Federal Information Processing Standards (FIPS) FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard, FIPS 204, Module-Lattice-Based Digital Signature Standard, and FIPS 205, Stateless Hash-Based Digital Signature Standard." Federal Register, August 14. <https://www.federalregister.gov/documents/2024/08/14/2024-17956/announcing-issuance-of-federal-information-processing-standards-fips-fips-203-module-lattice-based>.
- Preskill, John. 2018. "Quantum Computing in the NISQ Era and Beyond." *Quantum* 2 (August): 79. <https://doi.org/10.22331/q-2018-08-06-79>.

- Preskill, John. 2025. "Beyond NISQ: The Megaquop Machine." *ACM Transactions on Quantum Computing* 6 (3): 18:1-18:7. <https://doi.org/10.1145/3723153>.
- Putterman, Harald, Kyungjoo Noh, Connor T. Hann, et al. 2025. "Hardware-Efficient Quantum Error Correction via Concatenated Bosonic Qubits." *Nature* 638 (8052): 927–34. <https://doi.org/10.1038/s41586-025-08642-7>.
- Quantinuum Roadmap. "Quantinuum Roadmap." <https://www.quantinuum.com/blog/technical-perspective-by-the-end-of-the-decade-we-will-deliver-universal-fault-tolerant-quantum-computing>.
- QuEra Roadmap. "QuEra Roadmap." <https://www.quera.com/our-quantum-roadmap>.
- Sevilla, Jaime, and C. Jess Riedel. 2020. "Forecasting Timelines of Quantum Computing." *arXiv:2009.05045 [Quant-Ph]*, September 10. <http://arxiv.org/abs/2009.05045>.
- Wilhelm et al. 2025. *Status of Quantum Computer Development*. BSI. https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Publikationen/Studien/Quantencomputer/Entwicklungstand_QC_V_2_2.html?nn=916616.
- World Economic Forum. 2023. *Quantum Readiness Toolkit: Building a Quantum-Secure Economy*. World Economic Forum. <https://www.weforum.org/publications/quantum-readiness-toolkit-building-a-quantum-secure-economy/>.

A. Appendix

In this appendix, we provide information on various aspects of the reports, including a list of respondents, background on quantum computing, and details of our methodology.

A.1 List of respondents

A short description/bio that emphasizes the rationale for each respondent's inclusion is provided after the table. Respondents who have started participating in our surveys in 2019 are listed at the beginning and highlighted in grey.

#	Name	Institution	Country
1	Dorit Aharonov	Hebrew University of Jerusalem QEDMA Quantum Computing	ISR
2	Alexandre Blais	Institut quantique, Université de Sherbrooke	CAN
3	Ignacio Cirac	Max Planck Institute of Quantum Optics	DEU
4	Bill Coish	McGill University	CAN
5	David DiVincenzo	TU Delft	DEU
6	Martin Ekerå	KTH Royal Institute of Technology Swedish NCSA	SWE
7	Artur Ekert	University of Oxford	GBR/SGP
8	Daniel Gottesman	University of Maryland	USA
9	Andrea Morello	UNSW Sydney	AUS
10	Stephanie Simmons	Simon Fraser University Photonic Inc.	CAN
11	Peter Shor	Massachusetts Institute of Technology	USA
12	Frank Wilhelm-Mauch	Forschungszentrum Jülich GmbH	DEU
13	Sergio Boixo	Google Quantum AI	USA
14	Jens Eisert	Freie Universität Berlin	DEU
15	Joe Fitzsimons	Horizon Quantum	SGP
16	Jay Gambetta	IBM	USA
17	Yvonne Gao	National University of Singapore	SGP
18	Winfried Hensinger	University of Sussex Universal Quantum	GBR
19	Peter Knight	Imperial College Innovate UK	GBR
20	Yi-Kai Liu	US National Institute of Standards and Technology (NIST)	USA
21	Sabrina Maniscalco	University of Helsinki Algorithmiq	FIN
22	Nicolas Menicucci	RMIT University	AUS
23	Klaus Mølmer	Niels Bohr Institute, University of Copenhagen	DNK
24	Simone Severini	Lean FRO	USA
25	Lieven Vandersypen	QuTech, Delft University of Technology	NLD
26	Gregor Weihs	University of Innsbruck	AUT

Dorit Aharonov

A leader in quantum algorithms and complexity, and co-inventor of the quantum fault-tolerance threshold theorem.

Alexandre Blais

A leader in understanding how to control the quantum states of mesoscopic devices and applying the theoretical tools of quantum optics to mesoscopic systems, he has provided key theoretical contributions to the development of the field of circuit quantum electrodynamics with superconducting qubits.

Sergio Boixo

He is the Chief Scientist for Quantum Computer Theory at Google's Quantum Artificial Intelligence Lab. He is known for his work on quantum neural networks, quantum metrology, and was involved with the first-ever demonstration of quantum supremacy.

Ignacio Cirac

One of the pioneers of the field of quantum computing and quantum information theory. He established the theory at the basis of trapped-ion quantum computation. He devised new methods to efficiently study quantum systems on classical computers and to use controllable quantum systems (such as cold atoms) as quantum simulators.

Bill Coish

A theoretician working closely with experimentalists, he is a leading expert on solid-state quantum computing, including both spin-based and superconducting implementations.

David DiVincenzo

A pioneer in the field of quantum computing and quantum information theory. He formulated the "DiVincenzo criteria" that an effective physical implementation of quantum computing should satisfy.

Martin Ekerå

A leading cryptography researcher focusing on quantum computing algorithms for cryptanalysis, and on the development of post-quantum secure classical cryptographic schemes. He is the co-author of one of the most recent and influential estimates of the resources required by a realistic and imperfect quantum computer to break the RSA public-key encryption scheme.

Artur Ekert

A pioneer in the field of quantum information who works in quantum computation and communication. He invented entanglement-based quantum key distribution and was the founding director of the Centre for Quantum Technologies of Singapore.

Jens Eisert

He is recognized for his work in quantum information science and quantum many-body theory within condensed matter physics. His notable contributions include advances in entanglement theory, quantum computing research, protocols for quantum technologies, and the study of complex quantum systems.

Joe Fitzsimons

A leading theoretical physicist and CEO of Horizon Quantum Computing. He is renowned for his contributions to blind quantum computing. His current goal is to develop programming tools that simplify software development for quantum computers.

Jay Gambetta

He is the director of IBM Research and an IBM Fellow. He has led the team at IBM Thomas J Watson Research Center working to build a quantum computer. He is the recipient of the 2026 APS George E. Pake award for his expert leadership in the development and deployment of quantum computing technologies, as well as for making quantum computing accessible to researchers worldwide.

Yvonne Gao

Leads a group to develop modular quantum devices with superconducting quantum circuits. In 2019, she was named one of the Innovators Under 35 (Asia Pacific) by MIT Tech Review for her work in developing crucial building blocks for quantum computers.

Daniel Gottesman

A pioneer of quantum error correction, and inventor of the stabilizer formalism for quantum error correction. He is a Co-Director of the Joint Center for Quantum Information and Computer Science (QuICS).

Winfried Hensinger

He heads the Sussex Ion Quantum Technology Group and is the director of the Sussex Centre for Quantum Technologies. He is a co-founder, Chief Scientist and Chairman of Universal Quantum, a full-stack quantum computing company.

Sir Peter Knight

A physicist renowned for his pioneering research into quantum optics, including work on the nonclassical properties of light and the theoretical underpinnings of quantum computing. He spearheaded the UK National Quantum Technologies Programme. In recognition of his work in optical physics, he was knighted in the Queen's 2005 Birthday Honours List.

Yi-Kai Liu

He is a leader in research on quantum computation, quantum algorithms and complexity, quantum state tomography, and cryptography. He is a staff scientist in the Applied and Computational Mathematics Division at the National Institute of Standards and Technology (NIST), where he contributed to the definition of the first standards for Post-Quantum Cryptography.

Sabrina Maniscalco

A leading researcher in quantum open systems, quantum control, and quantum thermodynamics. She has made influential contributions to the theory of non-Markovian dynamics and their role in quantum technologies. She is the CEO and co-founder of Algorithmiq, a company developing quantum algorithms for life sciences.

Nicolas Menicucci

A leading researcher who contributed key results in the development of continuous-variable cluster

states, and who further focuses on foundational quantum information and quantum theory, in relation to relativity.

Klaus Mølmer

A pioneering physicist at the University of Aarhus, he has made outstanding and insightful contributions to theoretical quantum optics, quantum information science and quantum atom optics, including the development of novel computational methods to treat open systems in quantum mechanics and theoretical proposals for the quantum logic gates with trapped ions.

Andrea Morello

A leading experimentalist in the control of dynamics of spins in nanostructures. Prof Morello's group was the first in the world to achieve single-shot readout of an electron spin in silicon, and the coherent control of both the electron and the nuclear spin of a single donor.

Simone Severini

A leading researcher in quantum information and complex systems, particularly through the application of graph theory. He is currently Professor of Physics of Information at University College London, and served as initial Director of Quantum Computing at Amazon Web Services.

Stephanie Simmons

Co-leads the Silicon Quantum Technology Lab at Simon Fraser University and is an international expert on the experimental realization of spin qubits in silicon, and in interfacing them with photon qubits.

Peter Shor

The inventor of the efficient quantum algorithms for factoring and discrete logarithms that generated great interest in quantum computing, and a pioneer of quantum error correction.

Lieven Vandersypen

He demonstrated experimentally Shor's algorithm, famously factoring $15 = 3 \times 5$. Currently, He leads a group that pioneered quantum computing based on electron spins in semiconductor quantum dots. His current interests are to devise and demonstrate novel building blocks of a large-scale quantum dot spin qubit processor and to study many-body physics problems using quantum dot arrays.

Gregor Weihs

He is Professor of Photonics at the Institute for Experimental Physics at the University of Innsbruck, where he leads the Photonics group. His research in quantum optics and quantum information focuses on semiconductor nanostructures and on the foundations of quantum physics.

Frank Wilhelm-Mauch

A leading theoretician working closely with experimentalists, he focuses on modelling and controlling superconducting circuits. He is the director of the Peter Grünberg Institute for Quantum Computer Analytics.

A.2 Realizations of quantum computers

Physical realizations

The various physical implementations of quantum computers have advantages and disadvantages in relation to factors such as (but not limited to):

- *scalability*, that is, the possibility of building and controlling large quantum devices with more and more qubits using physical/engineering resources that grow in a manageable way;
- compatibility with—and ease of implementation of—different computational models;
- typical decoherence time (that is, for how long quantum features like superpositions remain preserved and can be exploited);
- speed and precision with which gates can be applied.

The following is a very high-level classification of some physical realizations:

- **Quantum optics**, meaning that information is stored and manipulated in states of light; this includes polarization states or photon-number states, and can be implemented also on-chip by using integrated optics.
- **Superconducting systems**, meaning that information is stored and manipulated in electric circuits that exploit the properties of superconducting materials.
- **Topological systems**, meaning that information is stored and manipulated in some topological properties—that is, properties that depend on ‘global’ (geometric) properties insensitive to ‘local’ changes—of quantum systems.
- **Ion traps**, meaning that information is stored and manipulated in properties of ions (atoms with non-vanishing total electric charge) that are confined by electro-magnetic fields.
- **Quantum spin systems**, meaning that information is stored and manipulated in the internal degree of freedom called *quantum spin*; such systems may be realized in silicon, like standard microchips are, or in less conventional systems, like diamonds with point defects known as nitrogen-vacancy (or NV, in short) centers.
- **Cold atoms**, where neutral atoms (rather than ions) are cooled down to close to absolute zero. While ions repel each other because of their electric charge, neutral atoms do not and can be trapped and arranged in very regular arrays via the use of laser beams that generate so-called optical lattices; the atoms can then be controlled all the way down to the level of individual sites in the lattice.

Models of computation

Besides many possible physical realizations of quantum computers, there are also various *models* of quantum computation. While many models are known to be computationally equivalent (that is, roughly speaking, they allow one to solve the same class of problems with similar efficiency), each model offers different insights into the design of algorithms or may be more suitable for a particular physical realization. One such model is the *circuit model*—or *gate model*—where transformations are

sequentially performed on single and multiple qubits (see Figure 31). From the perspective of analysing the quantum threat timeline, it is useful to focus on the circuit model as there is a well-articulated path to implementing impactful cryptanalytic attacks.

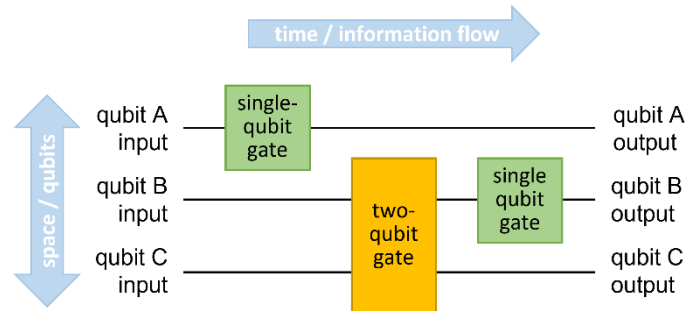


Figure 31 Illustration of the circuit/gate model for quantum computation. Each qubit corresponds to a horizontal line, so that multiple stacked lines illustrate many qubits. A qubit can be transformed individually by means of single-qubit gates, and two qubits can interact via a two-qubit gate. A given circuit transforms the initial input state of the qubits into their final output state, via the sequential action of said gates. The sequence of transformations is temporally ordered from left to right.

In the circuit model, to perform arbitrary computations it is enough to be able to realize a finite set of *universal gates* which can be combined to generate arbitrary transformations. Such a set necessarily includes at least one gate that let multiple qubits interact, typically two at a time.

Historically, the following criteria, which are part of a larger set of desiderata, and which were listed by DiVincenzo in (DiVincenzo 2000) and hence are known as *DiVincenzo's criteria*, have been considered essential requirements for any physical implementation of a quantum computer:

1. *A scalable physical system with well characterized qubits.*
2. *The ability to initialize the state of the qubits to a simple fiducial state.*
3. *Long relevant decoherence times, much longer than the gate operation time.*
4. *A “universal” set of quantum gates.*
5. *A qubit-specific measurement capability.*

Unfortunately, the implementation of a single- or multi-qubit transformation can never be exactly the intended one, as the parameters defining a transformation are continuous, and because of the inevitable noise/decoherence. The quality of a gate implementation can be quantified by some notion of *fidelity*: the larger the fidelity, the closer the implementation of a gate is to the ideal one. A related parameter is the physical *error rate* with which gates are applied. In a sense, this parameter is the ‘opposite’ of fidelity. When characterizing the gate quality of experimental realizations or when studying the theory of how to correct them, most research groups use either the fidelity or the error rate.

Error correction, fault tolerance, and logical qubits

Errors and imperfections in the manipulation of (quantum) information, as well as decoherence, may be reduced by improving the physical implementation, including qubit control, but they cannot be entirely eliminated. Nonetheless, reliable storage and processing of quantum can still be achieved by employing *error correction* schemes: *logical* qubits are encoded into multiple *physical* qubits, so that errors affecting the underlying physical qubits can be detected and corrected, and logical information be protected. Error correction can ultimately lead to *fault tolerance* (Nielsen and Chuang 2000): under reasonable assumptions, one can prove that, if the error rate of the underlying physical components is low enough—below the so-called *fault-tolerance threshold*—then it is possible to implement logical encodings for information and information processing that can be made arbitrarily reliable, at the cost of using a number of physical qubits that is potentially much larger than that of the encoded logical qubits, but that still scales in a manageable way, at least theoretically.

Some more details on such codes and techniques can be found below. Still, they are not as relevant as keeping in mind that quantum error correction and fault-tolerance do pave the way to digital quantum computers: in principle, quantum computing devices can be made as reliable as needed, once some “quality standard” and some scalability & integration of the underlying physical qubits are achieved. We provide information on some specific error-correcting codes to 1) facilitate the understanding of the expert opinions on the topic and 2) to make it clear that developing codes that enable fault tolerance, also considering their ease of realization and tailoring them to specific physical implementation, is an on-going and very important area of research. Most relevantly, improvements in error-correcting codes and/or in their hardware implementation may speed up the quantum threat timeline.

An important issue in error correction is the kind of errors that the adopted error-correction scheme/code can detect and correct.

In the case of classical bits, and excluding loss, the only possible type of error at the level of a single bit is the so-called *bit-flip*, which causes a 0 to turn into a 1, and vice versa. On the other hand, qubits can also undergo a so-called *phase-flip* error. Quantum codes can be designed and implemented that deal with just one of the two kinds of errors, but to protect quantum information both kinds need to be dealt with. Another important concept is that of *distance*, which roughly corresponds to the

number of physical qubits affected by an error that the error-correction scheme can handle. For example, the classical repetition code illustrated in Figure 32, using three physical bits to encode one logical bit, detects and corrects a single bit-flip error but would mishandle two bit-flips—confusing a logical 0 for a logical 1, and even introducing more physical errors upon correction. The special

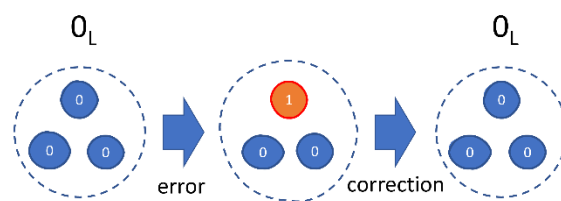


Figure 32 Example of classical information encoded logically. Several imperfect/error-prone physical bits (warped filled blue circles) are used to encode a logical 0, denoted O_L (dashed perfectly round circle), by means of a repetition code: O_L is encoded as 000 at the physical level. Errors can occur at the level of the physical bits, but they can be corrected, in this case by a simple majority-voting scheme, so that the logical bit is preserved. If the probability of a physical bit flipping is small enough, the probability of a logical bit being affected by an error—in this case, flipping from O_L to 1_L —is less than the probability of a physical flip. Quantum error correction can be seen as a generalization of classical error correction to protect quantum information; for example, a quantum code must preserve also (logical) superpositions of 0 and 1.

properties of quantum information prevent the use of simple repetition codes, but, in general, the ability to correct against more kinds of errors and against errors affecting more qubits leads to a higher number of physical qubits needed to encode a single logical qubit.

Examples of error correcting codes

Surface codes, which are an instance of so-called topological quantum error correcting codes (Kitaev 2003), are currently among the leading candidates for large-scale quantum error correction. The surface code (Fowler et al. 2012) allows for the detection and correction of errors on a two-dimensional array of nearest-neighbour coupled physical qubits via repeatedly measuring two types of so-called stabilizers generators. A single logical qubit is encoded into a square array of physical qubits. A classical error detection algorithm must be run at regular intervals (surface code cycle) to track the propagation of physical qubit errors and, ultimately, to prevent logical errors. Every surface code cycle involves some number of one- and two-qubit physical quantum gates, physical qubit measurements, and classical processing to detect and correct errors (i.e., decoding). Surface codes can provide logical qubits with lower overall error rates, at a price of increasing the number of physical qubits per logical qubit – that is, the size of the square array – and the cost of decoding.

Low-Density Parity Check (LDPC) codes have widespread use in the handling of classical information, as they have an essentially optimal scaling in terms of rate of encoding—the ratio between reliable logical bits and underlying faulty bits. Significant effort has recently been put into researching good *quantum LDPC codes*, which are characterized by the constraint that the number of underlying physical qubits involved in each error check and the number of checks each qubit is involved in are bounded by a constant (Breuckmann and Eberhardt 2021). One challenge with quantum LDPC codes is that the qubits used in the encoding and in the error correction, despite being “few”, may be far apart.

Bosonic codes constitute a major class of quantum error-correcting codes, alongside surface codes and quantum LDPC codes. Instead of encoding a logical qubit into many two-level systems, a bosonic code encodes it into one or a few harmonic oscillators—for example, a microwave cavity mode in a superconducting circuit or an optical mode. This encoding can be designed so that the main physical noise, such as single-photon loss or phase noise, is relatively straightforward to detect and correct. Notable examples include “cat codes” (Cochrane et al. 1999), where superpositions of coherent states encode logical 0 and 1, and Gottesman–Kitaev–Preskill (GKP) codes (Gottesman et al. 2001), where logical states are superpositions of position- and momentum-eigenstates arranged on a lattice in phase space. A key advantage of bosonic codes is their ability to embed some error protection directly into the physical degree of freedom, lowering the reliance on higher-level quantum error correction. However, bosonic codes also present challenges; for instance, preparing and manipulating cat states or GKP states require high-fidelity control and often complex hardware.

A.3 Survey Questions

Regarding the wording of the core questions, in general we wanted to minimize the chances that the respondents could interpret them very differently. For example, questions like “when will we have useful quantum computers?” or “is it likely that a quantum computer will break cryptography in 10 years?” would have been far too vague. Some could have assumed that a useful quantum computer could have just a few dozen physical qubits that can demonstrate some proof-of-concept speed-up over

currently known classical methods. Others could have assumed that a useful quantum computer will require thousands of logical qubits (and thus perhaps millions of physical qubits) and should be performing something of immediate commercial value. Even sticking to cryptographic applications, it is important to pose questions in the right way: a quantum computer breaking RSA-2048 in 10 years may be unlikely, but is it 49%, 10%, or 1% unlikely? Some of the above considerations and goals are in—perhaps, unavoidable—tension for some of the questions.

Given the scope of our survey, and the above general principles and considerations, we proceeded as follows:

- We kept the questions largely focused on the issue of the implementation of fault-tolerant quantum computers that would be able to run quantum algorithms posing an actual threat to cryptosystems.
- We sought a range of relevant perspectives. Already in 2019, we invited a select number of respondents with authoritative and profound insights. They provided a great variety of expertise on the most recent developments and the next steps needed towards the realization of fault-tolerant quantum computers. The same philosophy guided the selection of respondents in the subsequent surveys, including this one.
- Considering the quality of the pool of respondents, all very busy professionals and researchers, we kept the questions limited in number, so that the estimated time to complete the questionnaire was less than 30 minutes. In some cases, to secure responses to at least the major key question revolving around the quantum threat timeline, we gave the option to provide input about only such a key question.

NOTE: Given the latter flexibility, not all respondents have provided answers to all questions, some of which were optional to begin with.

- Given the inherent uncertainty in the progress towards realizing a quantum computer, we asked the respondents to indicate in a relatively coarse-grained fashion how likely something was to happen.
- In addition, for the main question, we offered the option to provide point estimates for the likelihood
- We did keep several of the questions at the basis of previous reports the same or very similar, so to be able to detect a change in opinions.
- On the other hand, we modified to some extent the set of questions from survey to survey, due to:
 - recent developments in the field (such as the efforts shifting towards quantum error correction and the realization of logical qubits) and in the economic, political, and social scenario;
 - the respondents' feedback from previous surveys;
 - the desire to seek opinions about other relevant aspects of the quantum threat timeline.
- For the non-free-form multiple-choice answers, we gave the possibility to leave more nuanced comments. This mitigated to some extent the issue of the experts potentially responding to the same questions under a different set of assumptions and allowed us to collect insightful opinions.

Here is a list of the main questions, grouped by questionnaire section.

Questions about “Implementations of quantum computing”

Q: Please indicate the potential of the following physical implementations for realizing a digital quantum computer with ~100 logical qubits in the next 5 years¹⁰.

Physical implementations listed: Superconducting Systems, Trapped Ions, Quantum Optics (including integrated photonics), Quantum spin systems in Silicon, Quantum spin systems not in Silicon, Topological Systems, Cold Atoms, Other

Options for answer: “Not promising”, “Some potential”, “Very promising”, “Lead candidate”, “No opinion”

Questions about “Timeframe estimates”

Q (key question): Please indicate how likely you estimate it is that a quantum computer able to factorize a 2048-bit number in less than 24 hours will be built within the next 5 years, 10 years, 15 years, 20 years, and 30 years.

Possible classification for each period of time:

1. Extremely unlikely (< 1% chance)
2. Very unlikely (< 5% chance)
3. Unlikely (< 30 % chance)
4. Neither likely nor unlikely (about 50% chance)
5. Likely (> 70 % chance)
6. Very likely (> 95% chance)
7. Extremely likely (> 99% chance)

Q: What do you consider the most promising scheme for fault-tolerance?

Q: Please indicate your likelihood estimates for useful commercial applications of available processors -- or of larger/less noisy processors but anyway not yet cryptographically-relevant -- going beyond proof-of-concept and/or promotional activities, within the indicated timeframes.

Timeframes: within 1 year, 3 years, 5 years, 10 years, and 15 years.

Possible classification for each period of time the same as for the key question.

Questions on “Other factors that may impact the quantum threat timeline”

Q: There might be research efforts going on “under the radar”, for example, by secret state-backed labs, by companies operating at least partially in stealth, by malicious private actors, etc. How significant do you think this kind of covert research component might be when it comes to affecting the timeline to breaking RSA-2048?

¹⁰ In previous surveys, we had asked about “the next 15 years” and, last year, “the next 10 years,” but this year, given the progress in the field and the aggressive roadmaps of some companies, we decided to inquire about a shorter timeframe.

Options: “Not significant”, “It might affect it by 1 year”, “It might affect it by 2 or 3 years”, “It might affect it by more than 3 years”, “No opinion”

Q: *You think that, over the next two years, the level of global investment (both by government and by industry) towards quantum computing will ...*

Options: “Significantly Increase”, “Increase”, “Stay about the same”, “Decrease”, “Significantly Decrease”, and “Prefer not to answer”

Q: *Which of the following is currently the front-runner in the "global race" to build a scalable fault-tolerant quantum computer?*

Options [multiple selection was possible]: China, Europe, North America, Other(s)

Q: *How likely are the following to be front-runners in the "global race" to build a scalable fault-tolerant quantum computer in five years?*

Each of “China”, “Europe”, “North America”, “Other(s)” could be assigned one evaluation among “Likely”, “Possibly”, “Unlikely”, “No Comment”

Q: *Do you expect that the following ethical concerns will influence the timeline for achieving a cryptographically relevant quantum computer, potentially through regulatory capture meant to mitigate them, where applicable?*

Listed concerns:

- “Breaking current encryption and endangering data privacy”
- “Only a few actors having quantum access (inequitable access)”
- “A quantum technology “arms race” increasing global tensions”
- “Lack of clarity/transparency about the capabilities of quantum computers (hype)”
- “Others(s) [please specify]”

Options for expectations:

- “Yes – the issue will significantly delay development (e.g. due to regulations or cautious research approaches)”
- “Possibly – there might be minor delays as the ethical issue is addressed”
- “No – technical progress will largely proceed regardless of this concern”
- “No opinion / Not sure”

Questions on “Current progress in the development of a cryptographically relevant quantum computer”

Q: *What has been the most significant recent (since the second half of 2024) achievement in the progress towards building a fault-tolerant quantum digital computer?*

Q: What do you consider to be the next essential step towards building a fault-tolerant quantum digital computer? (something that could reasonably be achieved by approximately Summer 2026)

Q: Please provide your opinion about the following aspects of quantum computing research as sources of substantial and potentially unexpected progress that may speed up the realization of cryptographically relevant quantum computers.

Aspects: “Quantum algorithms / quantum cryptanalysis”, “Quantum hardware”, “Error-correction schemes”, “Modular architecture”, “Compilation”, “Other (please indicate below)”

Potential levels: “High”, “Some”, “Low”, “No opinion”

Q: Please comment freely on the present and near-future status of development of quantum computers.

A.4 Responses and analysis

In this section of the Appendix, we provide some details on our methodology in handling and analyzing the responses.

Quantum factoring responses and analysis

We asked the respondents to provide an informative but rough estimate of the likelihood of the availability of a quantum computer able to factorize a 2048-bit number in less than 24 hours within a certain number of years. We provide here the raw aggregate counts of the responses.

LIKELIHOOD ESTIMATE	Within 5 years	Within 10 years	Within 15 years	Within 20 years	Within 30 years
Extremely unlikely (< 1% chance)	13	3	0	0	0
Very unlikely (< 5% chance)	5	4	3	0	0
Unlikely (< 30% chance)	6	6	5	2	1
Neither likely not unlikely (~ 50% chance)	1	7	3	6	3
Likely (> 70% chance)	1	4	9	6	5
Very likely (> 95% chance)	0	1	4	4	5
Extremely likely (> 99% chance)	0	1	2	8	12

To derive from the responses the cumulative probability distributions as shown in Section 4.2, we

assigned the following cumulative probabilities to each response, which are the largest and smallest ones compatible with the ranges among which the respondents could choose:

LIKELIHOOD ESTIMATE	OPTIMISTIC ASSIGNMENT	PESSIMISTIC ASSIGNMENT
Extremely likely (> 99% chance)	100%	99%
Very likely (> 95% chance)	99%	95%
Likely (> 70 % chance)	95%	70%
Neither likely nor unlikely (about 50% chance)	70%	30%
Unlikely (< 30 % chance)	30%	5%
Very unlikely (< 5% chance)	5%	1%
Extremely unlikely (< 1% chance)	1%	0%

The period option “More than 30 years, if ever” was implicit (not listed), and is trivially associated with a cumulative probability of 100%.

The cumulative probabilities of the experts have been simply averaged for both the optimistic and pessimistic assignments. For those respondents who provided also point estimates, we considered mean, median, and quartiles based on the numerical answers. Based on this additional information, here we provide also a version of the average likelihood estimates where point estimates are utilized where available (Figure 33).

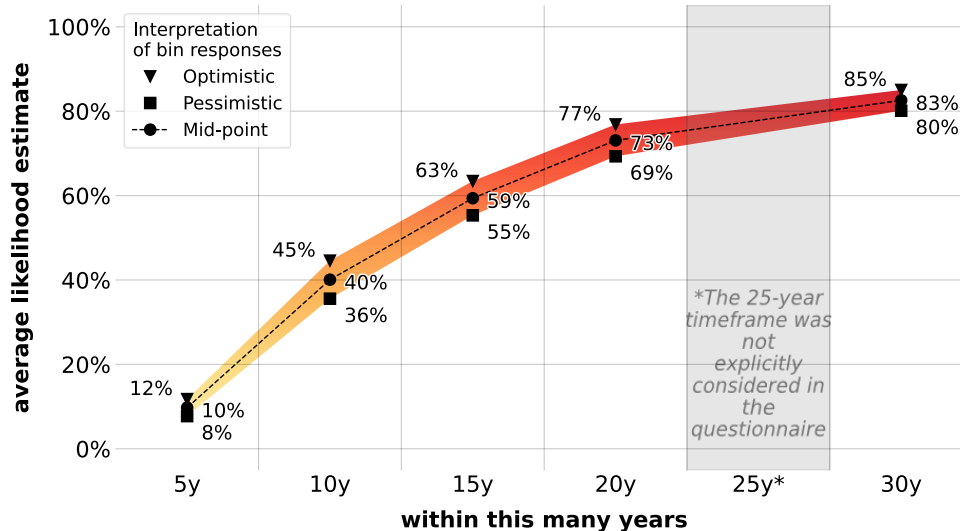
General considerations on the reliability of the experts’ estimates

We list here some considerations about factors that may influence the general reliability of the responses and/or lead to apparent changes in opinion trends:

- First and foremost, a general warning and an invitation to caution:
 - While the experts’ likelihood estimates provide insight into the quantum threat timeline, the results of our surveys must always be interpreted cautiously.
 - The experts who take part in our surveys are uniquely qualified to estimate the quantum threat timeline, but that does not imply that any of them can correctly indicate what is going to happen and when.
 - Both in this survey and in the previous ones, several experts themselves have explicitly admitted the difficulty of making reliable forecasts.



AVERAGE 2025 EXPERTS' ESTIMATES OF THE LIKELIHOOD OF A QUANTUM COMPUTER ABLE TO BREAK RSA-2048 IN 24 HOURS USING POINT ESTIMATES IF AVAILABLE



Note: Point estimates were used in the averages for those respondents that provided them, rather than the upper or lower bounds of their indicated likelihood intervals.

Figure 33 Average likelihood estimates, utilizing point estimates for those respondents who provided them.

- Considering averages does not necessarily provide the *best* possible estimates.
- When the pool of respondents changes from survey to survey, it may affect substantially the averages / the consensus.
- Statistically speaking, the number of respondents in our surveys is relatively small. Moreover, the time frame considered as well as the likelihood intervals constitute few, relatively coarse-grained bins. These factors may combine so that resulting estimates fluctuate noticeably from survey to survey, just because of few respondents answering slightly differently than they had done in the past. For example, if a respondent feels that a likelihood is around 25-35%, they might reasonably select “<30%” or “approximately 50%”, and “switch” choice from one survey to the next.
- The previous point is relevant even further when we adopt the approach of estimating likelihood ranges by interpreting optimistically or pessimistically the experts’ likelihood estimates; the reasons is that some of the likelihood ranges associated with some answers are larger than others.
- Especially from the perspective of someone working in quantum computing research and taking a survey like ours, the “time when a cryptographically relevant quantum computer will become available” is not a random value whose probability distribution is fixed. Our respondents are hard at work to make such a device become a reality, and the progress they achieve year after year is such that they are gaining a better understanding of the hurdles towards building it and of what needs to be done for circumventing them. This better understanding might increase confidence in the eventual realization of a quantum computer but might also allow them to better estimate how long it might take to overcome certain challenges. This corresponds to updating the above-mentioned distribution, for example making it more peaked some time in the future and, without contradiction, lower in the shorter term.

- Societal factors, including real or perceived issues related to the economy, may affect both the actual progress and perceptions/expectations about progress.

A.5 Glossary

This glossary provides short definitions of key terms and concepts related to quantum computing and cryptography as discussed in this report.

Superposition

The phenomenon in which the state of a quantum system can comprise alternatives that would be mutually exclusive in classical physics.

Qubit

The basic unit of quantum information. Unlike a classical bit that can be 0 or 1, a qubit can exist in a superposition of both states simultaneously, enabling quantum computers to process information in fundamentally different ways.

Logical Qubit

A robust unit of quantum information encoded across multiple physical qubits using error-correction techniques to mitigate noise and decoherence. Logical qubits are essential for fault-tolerant quantum computing.

Physical Qubit

An actual hardware implementation of a qubit, which is inherently imperfect and prone to errors. Multiple physical qubits are combined to form a logical qubit.

Entanglement

A uniquely quantum correlation in which two or more quantum systems become linked so that the state of one cannot be described independently of the others.

Decoherence

The process by which quantum systems lose their uniquely quantum properties, such as superposition, due to interactions with their surrounding environment.

Quantum Error Correction (QEC)

A set of techniques that protect quantum information by encoding it into logical qubits distributed across multiple physical qubits. QEC is critical for achieving fault tolerance in quantum computing.

Fault Tolerance

The ability of a quantum computer to continue operating correctly even when some components fail or experience errors, achieved through QEC and robust architectures.

Cryptographically Relevant Quantum Computer (CRQC)

A quantum computer powerful enough to break widely used cryptographic systems, such as RSA-2048, within practical timeframes.

Post-Quantum Cryptography (PQC)

Cryptographic algorithms designed to be secure against attacks by quantum computers. PQC includes lattice-based, hash-based, and other schemes standardized by NIST.

Harvest Now, Decrypt Later (HNDL)

An attack strategy where adversaries intercept and store encrypted data today, intending to decrypt it in the future when quantum computers become powerful enough to break current encryption.

Mosca Inequality

A principle stating that, if the time until the quantum threat is less than the sum of the data shelf-life and migration time, then sensitive data may become vulnerable to quantum-enabled breaches.

Shelf-Life Time

The number of years information must remain secure.

Migration Time

The time required to safely transition systems to quantum-safe cryptography.

Quantum Threat Time

The estimated time until adversaries have access to quantum computers capable of breaking current cryptographic systems.

Q-Day

The moment when quantum computers become sufficiently powerful to compromise current cryptographic standards, rendering many widely used encryption methods vulnerable.

NISQ (Noisy Intermediate-Scale Quantum)

Quantum devices with tens to hundreds of physical qubits, not yet fault-tolerant, but potentially useful for specific, specialized tasks.

FASQ (Fault-Tolerant Application-Scale Quantum)

A future class of quantum computers capable of running large-scale, error-corrected computations, including cryptanalysis.

Quantum LDPC (Low-Density Parity-Check) Codes

Quantum error-correcting codes that improve encoding efficiency, reducing the number of physical qubits needed per logical qubit. They may require relatively long-distance connectivity between qubits.

Surface Code

A quantum error-correcting code that encodes logical qubits into a two-dimensional array of nearest-neighbor coupled physical qubits.

Bosonic Codes

A class of quantum error-correcting codes that encode logical qubits into the states of harmonic

oscillators, such as microwave cavities or optical modes, rather than into multiple two-level systems (standard qubits).